



**PT. SALIPI NAGARI PERKASA**

Komp. Pejaten Mas Blok E No. 3 Kramat Watu Serang - Banten 42161

Telp. : 0254 - 231363 Fax.: 0254 - 231183

## **LAPORAN UJI MINYAK TRAFO**

**di**

**UNIT JASA PEMBANGKIT**

***JERANJANG***



Yang dilakukan pada tanggal 23 Oktober 2018

Untuk keadaan trafo bulan OKTOBER 2018

| <b><u>1.0 Table of Contents</u></b>                                                   | <b><u>Page</u></b> |
|---------------------------------------------------------------------------------------|--------------------|
| 1.0 Specific Information                                                              | 3                  |
| 2.0 Our Comments and Recommendations of the Test Results Analysis                     |                    |
| - <b>Transformer Unit MAIN TRAFO UNIT 1 (S/N: 100691)</b><br>Test Certificate 4       | 5                  |
| - <b>Transformer Unit TRAFO 1 ESP UNIT 1 (S/N: 10937)</b><br>Test Certificate 5       | 6                  |
| - <b>Transformer Unit TRAFO 2 ESP UNIT 1 (S/N: 10035)</b><br>Test Certificate 6       | 7                  |
| - <b>Transformer Unit TRAFO 3 ESP UNIT 1 (S/N: 10031)</b><br>Test Certificate 7       | 8                  |
| - <b>Transformer Unit TRAFO 4 ESP UNIT 1 (S/N: 10034)</b><br>Test Certificate 8       | 9                  |
| - <b>Transformer Unit MAIN TRAFO UNIT 3 (S/N: 1122170)</b><br>Test Certificate 9      | 10                 |
| - <b>Transformer Unit UAT 3 (S/N: 1035082)</b><br>Test Certificate 10                 | 11                 |
| - <b>Transformer Unit TRAFO 1 ESP UNIT 3 (S/N: 0073084208)</b><br>Test Certificate 11 | 12                 |
| - <b>Transformer Unit TRAFO 2 ESP UNIT 3 (S/N: 0070084208)</b><br>Test Certificate 12 | 13                 |
| - <b>Transformer Unit TRAFO 3 ESP UNIT 3 (S/N: 0072084208)</b><br>Test Certificate 13 | 14                 |
| - <b>Transformer Unit TRAFO 4 UNIT 3 (S/N: 0071084208)</b><br>Test Certificate 14     | 15                 |
| - <b>Transformer Unit ASH HANDLING-1 (S/N: 10820108)</b><br>Test Certificate 15       | 16                 |
| - <b>Transformer Unit ASH HANDLING-2 (S/N: 10820107)</b><br>Test Certificate 16       | 17                 |
| - <b>Transformer Unit TRAFO 1 ESP UNIT 2 (S/N: 10037)</b><br>Test Certificate 17      | 18                 |
| - <b>Transformer Unit TRAFO 2 ESP UNIT 2 (S/N: 10036)</b><br>Test Certificate 18      | 19                 |

**Page Disclaimer :**

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|     |                                                          |    |
|-----|----------------------------------------------------------|----|
| -   | <b>Transformer Unit TRAFO 3 ESP UNIT 2 (S/N: 10033)</b>  | 20 |
|     | Test Certificate 19                                      |    |
| -   | <b>Transformer Unit TRAFO 4 ESP UNIT 2 (S/N: 10032)</b>  | 21 |
|     | Test Certificate 20                                      |    |
| -   | <b>Transformer Unit MAIN TRAFO UNIT 2 (S/N: 1006902)</b> | 22 |
|     | Test Certificate 21                                      |    |
| 3.0 | <b>Summary of Recommendation</b>                         | 23 |

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**1.0 Specific Information:**

| No. | Unit No.           | Serial Number | TC | Test Requested                                              |
|-----|--------------------|---------------|----|-------------------------------------------------------------|
| 1.  | MAIN TRAFO UNIT 1  | 100691        | 4  | Acidity, Dielectric, Color, Visualization and Water Content |
| 2.  | TRAFO 1 ESP UNIT 1 | 10937         | 5  | Acidity, Dielectric, Color, Visualization and Water Content |
| 3.  | TRAFO 2 ESP UNIT 1 | 10035         | 6  | Acidity, Dielectric, Color, Visualization and Water Content |
| 4.  | TRAFO 3 ESP UNIT 1 | 10031         | 7  | Acidity, Dielectric, Color, Visualization and Water Content |
| 5.  | TRAFO 4 ESP UNIT 1 | 10034         | 8  | Acidity, Dielectric, Color, Visualization and Water Content |
| 6.  | MAIN TRAFO UNIT 3  | 1122170       | 9  | Acidity, Dielectric, Color, Visualization and Water Content |
| 7.  | UAT 3              | 1035082       | 10 | Acidity, Dielectric, Color, Visualization and Water Content |
| 8.  | TRAFO 1 ESP UNIT 3 | 0073084208    | 11 | Acidity, Dielectric, Color, Visualization and Water Content |
| 9.  | TRAFO 2 ESP UNIT 3 | 0070084208    | 12 | Acidity, Dielectric, Color, Visualization and Water Content |
| 10. | TRAFO 3 ESP UNIT 3 | 0072084208    | 13 | Acidity, Dielectric, Color, Visualization and Water Content |
| 11. | TRAFO 4 UNIT 3     | 0071084208    | 14 | Acidity, Dielectric, Color, Visualization and Water Content |
| 12. | ASH HANDLING-1     | 10820108      | 15 | Acidity, Dielectric, Color, Visualization and Water Content |
| 13. | ASH HANDLING-2     | 10820107      | 16 | Acidity, Dielectric, Color, Visualization and Water Content |
| 14. | TRAFO 1 ESP UNIT 2 | 10037         | 17 | Acidity, Dielectric, Color, Visualization and Water Content |
| 15. | TRAFO 2 ESP UNIT 2 | 10036         | 18 | Acidity, Dielectric, Color, Visualization and Water Content |
| 16. | TRAFO 3 ESP UNIT 2 | 10033         | 19 | Acidity, Dielectric, Color, Visualization and Water Content |
| 17. | TRAFO 4 ESP UNIT 2 | 10032         | 20 | Acidity, Dielectric, Color, Visualization and Water Content |

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| No. | Unit No.          | Serial Number | TC | Test Requested                                              |
|-----|-------------------|---------------|----|-------------------------------------------------------------|
| 18. | MAIN TRAFO UNIT 2 | 1006902       | 21 | Acidity, Dielectric, Color, Visualization and Water Content |

**Note:**

\* OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

\*\* Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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**2.0 Our Comments and Recommendations of The Test Results Analysis:**

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

| No. | Unit No.          | Serial Number | TC |
|-----|-------------------|---------------|----|
| 1.  | MAIN TRAFO UNIT 1 | 100691        | 4  |

**A) Analysis**

- **Acidity** of 0.014mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 83kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 8ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.36% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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# CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

4

DATE PRINTED 13/11/2018

ID# - 601669

**CUST - PT IP UJP PLTU  
JERANJANG**

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **MAIN TRAFU UNIT 1**

LOCATION - OUTDOOR  
OTHER -

## VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | OFFLINE | 48          |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | ONLINE  | 60          | 64         | NORMAL    | FAIR  | GOOD   | NONE  |
| 27/11/2017 | ONLINE  |             | 51         |           |       |        |       |
| 06/12/2017 | OFFLINE | 50          | 48         |           |       |        |       |
| 22/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  | 50          | 55         |           |       |        |       |
| 24/07/2018 | ONLINE  |             |            |           |       | GOOD   |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | ONLINE  |             |            |           |       |        |       |
| 23/10/2018 | ONLINE  |             |            |           |       |        |       |

## LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.014 AC                        |                            | 82 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.014 AC                        |                            | 79 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 27/11/2017 | ROUTINE  | 0.014 AC                        |                            | 53 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.011 AC                        | 40.2 AC                    | 87 AC                   | 0.5 AC               | 0.8745 AC             | CLEAR AC             |                          |
| 22/05/2018 | ROUTINE  | 0.011 AC                        |                            | 77 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.014 AC                        | 41.9 AC                    | 69 AC                   | 0.5 AC               | 0.8716 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.014 AC                        |                            | 86 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.017 AC                        |                            | 90 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.014 AC                        |                            | 96 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.014 AC                        |                            | 83 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

## DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 147      | 9471   | 43278    | 32      | 655                | 2728              | 4      | 5        | ND        | 843                  | 56320        |
| 06/12/2017 | 183      | 12895  | 43794    | 39      | 715                | 3166              | 5      | 5        | ND        | 947                  | 60802        |
| 28/06/2018 | 176      | 11044  | 46776    | 41      | 758                | 3393              | 5      | 6        | ND        | 986                  | 62199        |

## RECOMMENDATION:

NO FURTHER GENERATION OF HYDROGEN IS NOTED. HOWEVER, INCREASING PATTERN OF CARBON OXIDES ARE DETECTED.

## NAMEPLATE DATA

|               |                 |
|---------------|-----------------|
| MFG           | - WOLONG        |
| DATE MFG      | - 2011          |
| MAIN TANK S/N | - <b>100691</b> |
| KVA           | - 40000         |
| PRI           | - 165000 V      |
| SEC           | - 6300 V        |
| KILOGRAMS     | - 81000         |
| EQUIP TYPE    | - TRANSFORMER   |
| TRANS CLASS   | - OFAF          |
| IMPEDENCE     | - 13.1%         |
| PHASE/CYCLE   | - 3/50 Hz       |
| LIQUID TYPE   | - OIL           |
| LITRES LIQUID | - 30857         |

|                   |   |
|-------------------|---|
| LTC MFG           | - |
| LTC S/N           | - |
| LTC FAB NO        | - |
| LTC MODEL         | - |
| LTC LITRES LIQUID | - |
| LTC ENERGIZED     | - |

## ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - YES |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - YES |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - YES |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

## WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 36     | 8   | 0.57  | A     |
| 30/10/2017 | 44     | 10  | 0.45  | A     |
| 27/11/2017 | 42     | 8   | 0.40  | A     |
| 06/12/2017 | 38     | 7   | 0.46  | A     |
| 22/05/2018 | 30     | 10  | 1.05  | B     |
| 28/06/2018 | 41     | 7   | 0.35  | A     |
| 24/07/2018 | 32     | 6   | 0.57  | A     |
| 25/08/2018 | 30     | 6   | 0.63  | A     |
| 25/09/2018 | 38     | 7   | 0.46  | A     |
| 23/10/2018 | 44     | 8   | 0.36  | A     |

## FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - MAIN TRAF0 UNIT 1

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |               |
|-------------------|---------------|
| MFG               | - WOLONG      |
| DATE MFG          | - 2011        |
| MAIN TANK S/N     | - 100691      |
| KVA               | - 40000       |
| PRI               | - 165000 V    |
| SEC               | - 6300 V      |
| KILOGRAMS         | - 81000       |
| EQUIP TYPE        | - TRANSFORMER |
| TRANS CLASS       | - OFAF        |
| IMPEDENCE         | - 13.1%       |
| PHASE/CYCLE       | - 3/50 Hz     |
| LIQUID TYPE       | - OIL         |
| LITRES LIQUID     | - 30857       |
| LTC MFG           | -             |
| LTC S/N           | -             |
| LTC FAB NO        | -             |
| LTC MODEL         | -             |
| LTC LITRES LIQUID | -             |
| LTC ENERGIZED     | -             |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - YES |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - YES |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - YES |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF  
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | 0.02       | 0.01  | 0.01   |
| 06/12/2017 | ND         | 0.01  | 0.04   |
| 28/06/2018 | ND         | ND    | 0.04   |

LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 0.140 AC |
| 06/12/2017 | 0.000 AC | 0.040 AC |
| 28/06/2018 | 0.000 AC | 0.140 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.24 AC    |
| 06/12/2017 | 0.27 AC    |
| 28/06/2018 | 0.24 AC    |

CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE    |
|------------|---------|-------|---------------|
| 04/07/2017 | NON-COR | 3A    | MAGENTA       |
| 06/12/2017 | NON-COR | 2C    | LAVENDER BLUE |
| 28/06/2018 | NON-COR | 3A    | MAGENTA       |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 2.  | TRAFO 1 ESP UNIT 1 | 10937         | 5  |

**A) Analysis**

- **Acidity** of 0.020mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 49kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 24ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.32% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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DATE PRINTED 12/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | OFFLINE | 35          |            | NORMAL    |       | GOOD   | NONE  |
| 30/10/2017 | OFFLINE | 50          |            | NORMAL    |       | GOOD   | NONE  |
| 27/11/2017 | ONLINE  |             |            |           |       |        |       |
| 06/12/2017 | OFFLINE |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | ONLINE  |             |            |           |       |        |       |
| 23/10/2018 | ONLINE  |             |            |           |       |        |       |

NAMEPLATE DATA

|               |                |
|---------------|----------------|
| MFG           | - FUJIAN       |
| DATE MFG      | - 2010         |
| MAIN TANK S/N | - <b>10937</b> |
| KVA           | -              |
| PRI           | - 72000 V      |
| SEC           | - 380 V        |
| KILOGRAMS     | - 1000         |
| EQUIP TYPE    | - TRANSFORMER  |
| TRANS CLASS   | - ONAN         |
| IMPEDENCE     | -              |
| PHASE/CYCLE   | - 3/50 Hz      |
| LIQUID TYPE   | - OIL          |
| LITRES LIQUID | - 630          |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.059 QU                        |                            | 58 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.059 QU                        |                            | 23 UN                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 27/11/2017 | ROUTINE  | 0.062 QU                        |                            | 25 QU                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.062 QU                        | 25.0 UN                    | 48 AC                   | 1.5 AC               | 0.8583 AC             | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.065 QU                        |                            | 38 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.067 QU                        | 25.9 UN                    | 32 AC                   | 1.5 AC               | 0.8560 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.073 QU                        |                            | 38 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.020 AC                        |                            | 71 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.017 AC                        |                            | 63 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.020 AC                        |                            | 49 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 35       | 20494  | 67487    | 4       | 69                 | 1143              | ND     | 1        | ND        | 109                  | 89233        |
| 06/12/2017 | 39       | 19933  | 67198    | 4       | 83                 | 1261              | ND     | 1        | ND        | 127                  | 88519        |
| 28/06/2018 | 44       | 20039  | 64529    | 4       | 119                | 1696              | ND     | 2        | ND        | 169                  | 86433        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 32     | 19  | 1.80  | D     |
| 30/10/2017 | 42     | 32  | 1.60  | C     |
| 27/11/2017 | 41     | 33  | 1.65  | C     |
| 06/12/2017 | 30     | 18  | 1.89  | D     |
| 21/05/2018 | 31     | 33  | 3.13  | E     |
| 28/06/2018 | 32     | 30  | 2.85  | E     |
| 24/07/2018 | 29     | 19  | 2.00  | D     |
| 25/08/2018 | 32     | 12  | 1.14  | B     |
| 25/09/2018 | 35     | 18  | 1.28  | B     |
| 23/10/2018 | 40     | 24  | 1.32  | B     |

FIELD SERVICE

DATE SERVICE  
01/08/2018 CHANGED OIL

# CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

5

DATE PRINTED 12/11/2018

ID# - 601669

**CUST - PT IP UJP PLTU  
JERANJANG**

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

## NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10937</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

## ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

## FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | 95   | ND   | ND   | ND   | 95    |
| 06/12/2017 | ND   | 75   | ND   | ND   | ND   | 75    |
| 28/06/2018 | ND   | 85   | ND   | ND   | ND   | 85    |

RECOMMENDATION:  
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

## METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | ND         | 0.01  | 0.01   |
| 06/12/2017 | ND         | ND    | ND     |
| 28/06/2018 | ND         | ND    | 0.03   |

## LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 1.870 AC |
| 06/12/2017 | 0.000 AC | 0.570 AC |
| 28/06/2018 | 0.000 AC | 0.680 AC |

## INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | ND UN      |
| 06/12/2017 | ND UN      |
| 28/06/2018 | ND UN      |

## CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE   |
|------------|---------|-------|--------------|
| 04/07/2017 | NON-COR | 1B    | DARK ORANGE  |
| 06/12/2017 | NON-COR | 1A    | LIGHT ORANGE |
| 28/06/2018 | NON-COR | 1B    | DARK ORANGE  |

## PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

## FIELD SERVICE

DATE SERVICE  
01/08/2018 CHANGED OIL

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 3.  | TRAFO 2 ESP UNIT 1 | 10035         | 6  |

**A) Analysis**

- **Acidity** of 0.020mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 71kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 14ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.15% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

**Page Disclaimer :**

*All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein*

DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | OFFLINE | 40          |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | OFFLINE | 44          |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 27/11/2017 | ONLINE  |             |            |           |       |        |       |
| 06/12/2017 | OFFLINE |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | LOW       |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | OFFLINE |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10035</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.048 AC                        |                            | 59 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.051 QU                        |                            | 62 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 27/11/2017 | ROUTINE  | 0.051 QU                        |                            | 50 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.053 QU                        | 27.3 UN                    | 51 AC                   | 1.5 AC               | 0.8589 AC             | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.056 QU                        |                            | 45 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.059 QU                        | 27.6 UN                    | 55 AC                   | 1.5 AC               | 0.8570 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.062 QU                        |                            | 47 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.022 AC                        |                            | 56 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.020 AC                        |                            | 67 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.020 AC                        |                            | 71 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 27       | 18639  | 66781    | 3       | 64                 | 1106              | ND     | 1        | ND        | 95                   | 86621        |
| 06/12/2017 | 20       | 19360  | 61590    | 3       | 45                 | 728               | ND     | 1        | ND        | 69                   | 81747        |
| 28/06/2018 | 22       | 22494  | 64443    | 4       | 57                 | 789               | ND     | ND       | ND        | 83                   | 87809        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 31     | 19  | 1.80  | D     |
| 30/10/2017 | 35     | 17  | 1.21  | B     |
| 27/11/2017 | 35     | 18  | 1.28  | B     |
| 06/12/2017 | 30     | 17  | 1.78  | D     |
| 21/05/2018 | 31     | 20  | 1.90  | D     |
| 28/06/2018 | 32     | 17  | 1.62  | C     |
| 24/07/2018 | 30     | 15  | 1.58  | C     |
| 25/08/2018 | 31     | 13  | 1.24  | B     |
| 25/09/2018 | 31     | 11  | 1.04  | B     |
| 23/10/2018 | 33     | 14  | 1.15  | B     |

FIELD SERVICE

DATE SERVICE  
01/08/2018 CHANGED OIL

# CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

6

DATE PRINTED 13/11/2018

ID# - 601669

**CUST - PT IP UJP PLTU  
JERANJANG**

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

## NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10035</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

## ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

## FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | 62   | ND   | ND   | ND   | 62    |
| 06/12/2017 | ND   | 67   | ND   | ND   | ND   | 67    |
| 28/06/2018 | ND   | 84   | ND   | ND   | ND   | 84    |

RECOMMENDATION:  
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

## METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | ND         | 0.01  | ND     |
| 06/12/2017 | ND         | ND    | 0.02   |
| 28/06/2018 | ND         | ND    | 0.02   |

## LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 1.900 AC |
| 06/12/2017 | 0.000 AC | 0.980 AC |
| 28/06/2018 | 0.000 AC | 1.310 AC |

## INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.02 UN    |
| 06/12/2017 | 0.02 UN    |
| 28/06/2018 | ND UN      |

## CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE   |
|------------|---------|-------|--------------|
| 04/07/2017 | NON-COR | 1A    | LIGHT ORANGE |
| 06/12/2017 | NON-COR | 1B    | DARK ORANGE  |
| 28/06/2018 | NON-COR | 1B    | DARK ORANGE  |

## PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

## FIELD SERVICE

DATE SERVICE  
01/08/2018 CHANGED OIL

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 4.  | TRAFO 3 ESP UNIT 1 | 10031         | 7  |

**A) Analysis**

- **Acidity** of 0.056mgKOH/gm is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 44kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 20ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 1.42% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates the presence of high water content in the oil.

**B) Recommendation**

- Based on previous oil analysis, the IFT of the oil is unfavorable. Together with the high level of Acidity and Water Content, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

**Page Disclaimer :**

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DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | OFFLINE | 38          |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | OFFLINE | 44          |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 27/11/2017 | ONLINE  |             |            |           |       |        |       |
| 06/12/2017 | OFFLINE |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | OFFLINE |             |            |           |       |        |       |
| 23/10/2018 | ONLINE  |             |            |           |       |        |       |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.045 AC                        |                            | 72 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.048 AC                        |                            | 72 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 27/11/2017 | ROUTINE  | 0.051 QU                        |                            | 62 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.053 QU                        | 28.3 QU                    | 57 AC                   | 1.5 AC               | 0.8624 AC             | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.056 QU                        |                            | 48 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.053 QU                        | 29.7 QU                    | 62 AC                   | 1.5 AC               | 0.8605 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.051 QU                        |                            | 64 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.056 QU                        |                            | 57 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.051 QU                        |                            | 46 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.056 QU                        |                            | 44 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 32       | 15273  | 65111    | 5       | 95                 | 1002              | ND     | 2        | ND        | 134                  | 81520        |
| 06/12/2017 | 7        | 14730  | 65738    | 3       | 42                 | 617               | ND     | ND       | ND        | 52                   | 81137        |
| 28/06/2018 | 9        | 17958  | 66007    | 4       | 46                 | 781               | ND     | 1        | ND        | 60                   | 84806        |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 32     | 14  | 1.33  | B     |
| 30/10/2017 | 37     | 16  | 1.04  | B     |
| 27/11/2017 | 34     | 17  | 1.39  | C     |
| 06/12/2017 | 30     | 16  | 1.68  | C     |
| 21/05/2018 | 31     | 21  | 2.00  | D     |
| 28/06/2018 | 30     | 17  | 1.78  | D     |
| 24/07/2018 | 30     | 17  | 1.78  | D     |
| 25/08/2018 | 30     | 15  | 1.58  | C     |
| 25/09/2018 | 32     | 21  | 2.00  | D     |
| 23/10/2018 | 36     | 20  | 1.42  | C     |

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10031</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

FIELD SERVICE

|      |         |
|------|---------|
| DATE | SERVICE |
|------|---------|

# CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

7

DATE PRINTED 13/11/2018

ID# - 601669

**CUST - PT IP UJP PLTU  
JERANJANG**

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

## NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10031</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

## ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

## FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | 32   | ND   | ND   | ND   | 32    |
| 06/12/2017 | ND   | 14   | ND   | ND   | ND   | 14    |
| 28/06/2018 | ND   | 20   | ND   | ND   | ND   | 20    |

RECOMMENDATION:  
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A  
HEALTHY SOLID INSULATION.

## METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | 0.01       | 0.01  | 0.01   |
| 06/12/2017 | ND         | ND    | ND     |
| 28/06/2018 | ND         | ND    | 0.11   |

## LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 1.700 AC |
| 06/12/2017 | 0.000 AC | 0.630 AC |
| 28/06/2018 | 0.000 AC | 1.030 AC |

## INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.01 UN    |
| 06/12/2017 | 0.01 UN    |
| 28/06/2018 | ND UN      |

## CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE    |
|------------|---------|-------|---------------|
| 04/07/2017 | NON-COR | 1A    | LIGHT ORANGE  |
| 06/12/2017 | NON-COR | 2C    | LAVENDER BLUE |
| 28/06/2018 | NON-COR | 1B    | DARK ORANGE   |

## PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

## FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 5.  | TRAFO 4 ESP UNIT 1 | 10034         | 8  |

**A) Analysis**

- **Acidity** of 0.014mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 52kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 21ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 1.05% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates the presence of high water content in the oil.

**B) Recommendation**

- Based on previous oil analysis, the IFT of the oil is unfavorable. Together with the high level Water Content, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

**Page Disclaimer :**

*All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein*

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | OFFLINE | 40          |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | OFFLINE | 44          |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 27/11/2017 | ONLINE  |             |            |           |       |        |       |
| 06/12/2017 | OFFLINE |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | LOW       | FAIR  | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | OFFLINE |             |            |           |       |        |       |
| 23/10/2018 | ONLINE  |             |            |           |       |        |       |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.031 AC                        |                            | 53 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.028 AC                        |                            | 48 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 27/11/2017 | ROUTINE  | 0.034 AC                        |                            | 41 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.031 AC                        | 27.2 UN                    | 41 AC                   | 1.5 AC               | 0.8645 AC             | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.034 AC                        |                            | 37 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.037 AC                        | 27.5 UN                    | 55 AC                   | 1.5 AC               | 0.8635 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.039 AC                        |                            | 50 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.017 AC                        |                            | 61 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.014 AC                        |                            | 45 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.014 AC                        |                            | 52 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 13       | 12265  | 66770    | 5       | 172                | 1206              | ND     | 8        | ND        | 198                  | 80439        |
| 06/12/2017 | 8        | 13331  | 63527    | 3       | 70                 | 720               | ND     | 1        | ND        | 82                   | 77660        |
| 28/06/2018 | 9        | 17962  | 64248    | 4       | 75                 | 1054              | ND     | 2        | ND        | 90                   | 83354        |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 33     | 15  | 1.23  | B     |
| 30/10/2017 | 39     | 18  | 0.99  | B     |
| 27/11/2017 | 37     | 30  | 1.95  | D     |
| 06/12/2017 | 30     | 18  | 1.89  | D     |
| 21/05/2018 | 31     | 23  | 2.18  | D     |
| 28/06/2018 | 30     | 20  | 2.10  | D     |
| 24/07/2018 | 31     | 19  | 1.80  | D     |
| 25/08/2018 | 30     | 14  | 1.47  | C     |
| 25/09/2018 | 30     | 20  | 2.10  | D     |
| 23/10/2018 | 41     | 21  | 1.05  | B     |

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10034</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

FIELD SERVICE

|            |             |
|------------|-------------|
| DATE       | SERVICE     |
| 01/08/2018 | CHANGED OIL |

# CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

8

DATE PRINTED 13/11/2018

ID# - 601669

**CUST - PT IP UJP PLTU  
JERANJANG**

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

## NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10034</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

## ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

## FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | 18   | ND   | ND   | ND   | 18    |
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | 21   | ND   | ND   | ND   | 21    |

RECOMMENDATION:  
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A  
HEALTHY SOLID INSULATION.

## METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | 0.05       | 0.02  | 0.01   |
| 06/12/2017 | ND         | ND    | 0.02   |
| 28/06/2018 | ND         | ND    | 0.05   |

## LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.180 QU | 6.680 UN |
| 06/12/2017 | 0.010 AC | 2.180 AC |
| 28/06/2018 | 0.060 AC | 2.850 AC |

## INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.08 UN    |
| 06/12/2017 | 0.06 UN    |
| 28/06/2018 | 0.02 UN    |

## CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE    |
|------------|---------|-------|---------------|
| 04/07/2017 | NON-COR | 2C    | LAVENDER BLUE |
| 06/12/2017 | NON-COR | 2C    | LAVENDER BLUE |
| 28/06/2018 | NON-COR | 3A    | MAGENTA       |

## PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

## FIELD SERVICE

DATE SERVICE  
01/08/2018 CHANGED OIL

| No. | Unit No.          | Serial Number | TC |
|-----|-------------------|---------------|----|
| 6.  | MAIN TRAF0 UNIT 3 | 1122170       | 9  |

**A) Analysis**

- **Acidity** of 0.017mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 78kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 8ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.76% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - MAIN TRAFU UNIT 3

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | ONLINE  | 58          | 68         | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | OFFLINE | 32          | 32         | NORMAL    | FAIR  | GOOD   | NONE  |
| 27/11/2017 | ONLINE  | 46          | 50         |           |       |        |       |
| 06/12/2017 | ONLINE  | 57          | 48         |           |       |        |       |
| 22/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  | 48          | 50         | NORMAL    |       |        |       |
| 24/07/2018 | ONLINE  |             |            |           |       |        |       |
| 25/08/2018 | ONLINE  |             |            |           |       |        |       |
| 25/09/2018 | ONLINE  |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

NAMEPLATE DATA

|               |               |
|---------------|---------------|
| MFG           | - SHANDONG    |
| DATE MFG      | - 2010        |
| MAIN TANK S/N | - 1122170     |
| KVA           | - 40000       |
| PRI           | - 165000 V    |
| SEC           | - 6300 V      |
| KILOGRAMS     | - 73920       |
| EQUIP TYPE    | - TRANSFORMER |
| TRANS CLASS   | - ONAF        |
| IMPEDENCE     | - 13.79%      |
| PHASE/CYCLE   | - 3/50 Hz     |
| LIQUID TYPE   | - OIL         |
| LITRES LIQUID | - 26708       |

|                   |   |
|-------------------|---|
| LTC MFG           | - |
| LTC S/N           | - |
| LTC FAB NO        | - |
| LTC MODEL         | - |
| LTC LITRES LIQUID | - |
| LTC ENERGIZED     | - |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.014 AC                        |                            | 86 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.014 AC                        |                            | 88 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 27/11/2017 | ROUTINE  | 0.014 AC                        |                            | 87 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.014 AC                        | 41.5 AC                    | 84 AC                   | 0.5 AC               | 0.8832 AC             | CLEAR AC             |                          |
| 22/05/2018 | ROUTINE  | 0.014 AC                        |                            | 58 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.014 AC                        | 41.0 AC                    | 85 AC                   | 0.5 AC               | 0.8805 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.014 AC                        |                            | 77 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.017 AC                        |                            | 82 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.017 AC                        |                            | 83 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.017 AC                        |                            | 78 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON MONOXIDE | CARBON DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL COMBUSTIBLE | TOTAL GAS |
|------------|----------|--------|----------|---------|-----------------|----------------|--------|----------|-----------|-------------------|-----------|
| 04/07/2017 | 24       | 10131  | 44084    | 11      | 641             | 2250           | 1      | 1        | ND        | 678               | 57143     |
| 06/12/2017 | 39       | 14023  | 43226    | 11      | 668             | 2456           | ND     | 1        | ND        | 719               | 60424     |
| 28/06/2018 | 18       | 11431  | 48283    | 13      | 700             | 2512           | 1      | 1        | ND        | 733               | 62959     |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - YES |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - YES |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - YES |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

RECOMMENDATION:  
CARBON MONOXIDE AND CARBON DIOXIDE LEVEL ARE HIGH. OTHER GASES LEVELS ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 42     | 10  | 0.50  | A     |
| 30/10/2017 | 28     | 9   | 1.06  | B     |
| 27/11/2017 | 35     | 8   | 0.57  | A     |
| 06/12/2017 | 37     | 9   | 0.58  | A     |
| 22/05/2018 | 31     | 11  | 1.04  | B     |
| 28/06/2018 | 36     | 8   | 0.57  | A     |
| 24/07/2018 | 33     | 8   | 0.66  | A     |
| 25/08/2018 | 32     | 7   | 0.67  | A     |
| 25/09/2018 | 36     | 6   | 0.43  | A     |
| 23/10/2018 | 32     | 8   | 0.76  | A     |

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - MAIN TRAF0 UNIT 3

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |               |
|-------------------|---------------|
| MFG               | - SHANDONG    |
| DATE MFG          | - 2010        |
| MAIN TANK S/N     | - 1122170     |
| KVA               | - 40000       |
| PRI               | - 165000 V    |
| SEC               | - 6300 V      |
| KILOGRAMS         | - 73920       |
| EQUIP TYPE        | - TRANSFORMER |
| TRANS CLASS       | - ONAF        |
| IMPEDENCE         | - 13.79%      |
| PHASE/CYCLE       | - 3/50 Hz     |
| LIQUID TYPE       | - OIL         |
| LITRES LIQUID     | - 26708       |
| LTC MFG           | -             |
| LTC S/N           | -             |
| LTC FAB NO        | -             |
| LTC MODEL         | -             |
| LTC LITRES LIQUID | -             |
| LTC ENERGIZED     | -             |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - YES |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - YES |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - YES |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF  
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | 0.01       | 0.01  | 0.02   |
| 06/12/2017 | 0.01       | ND    | 0.03   |
| 28/06/2018 | ND         | 0.01  | 0.06   |

LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 0.130 AC |
| 06/12/2017 | 0.000 AC | 0.070 AC |
| 28/06/2018 | 0.000 AC | 0.110 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.22 AC    |
| 06/12/2017 | 0.22 AC    |
| 28/06/2018 | 0.19 QU    |

CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE    |
|------------|---------|-------|---------------|
| 04/07/2017 | NON-COR | 3A    | MAGENTA       |
| 06/12/2017 | NON-COR | 2C    | LAVENDER BLUE |
| 28/06/2018 | NON-COR | 3A    | MAGENTA       |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

| No. | Unit No. | Serial Number | TC |
|-----|----------|---------------|----|
| 7.  | UAT 3    | 1035082       | 10 |

**A) Analysis**

- **Acidity** of 0.020mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 69kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 8ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.66% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

**Page Disclaimer :**

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DATE PRINTED 13/11/2018

ID# - 601669

**CUST - PT IP UJP PLTU  
JERANJANG**SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **UAT 3**LOCATION - OUTDOOR  
OTHER -

## VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | ONLINE  |             | 58         | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | OFFLINE |             | 44         | NORMAL    | FAIR  | GOOD   | NONE  |
| 27/11/2017 | ONLINE  |             | 60         |           |       |        |       |
| 06/12/2017 | ONLINE  |             | 60         |           |       |        |       |
| 22/05/2018 | ONLINE  |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             | 50         |           |       |        |       |
| 24/07/2018 | ONLINE  |             |            |           |       |        |       |
| 25/08/2018 | ONLINE  |             |            |           |       |        |       |
| 25/09/2018 | ONLINE  |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

## NAMEPLATE DATA

|               |                  |
|---------------|------------------|
| MFG           | - SHANDONG       |
| DATE MFG      | - 2010           |
| MAIN TANK S/N | - <b>1035082</b> |
| KVA           | - 6300           |
| PRI           | - 6300 V         |
| SEC           | - 6300 V         |
| KILOGRAMS     | - 14600          |
| EQUIP TYPE    | - TRANSFORMER    |
| TRANS CLASS   | - ONAN           |
| IMPEDENCE     | - 5.78%          |
| PHASE/CYCLE   | - 3/50 Hz        |
| LIQUID TYPE   | - OIL            |
| LITRES LIQUID | - 3749           |

|                   |   |
|-------------------|---|
| LTC MFG           | - |
| LTC S/N           | - |
| LTC FAB NO        | - |
| LTC MODEL         | - |
| LTC LITRES LIQUID | - |
| LTC ENERGIZED     | - |

## LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.020 AC                        |                            | 80 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.022 AC                        |                            | 86 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 27/11/2017 | ROUTINE  | 0.022 AC                        |                            | 86 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.020 AC                        | 37.6 AC                    | 86 AC                   | 0.5 AC               | 0.8820 AC             | CLEAR AC             |                          |
| 22/05/2018 | ROUTINE  | 0.020 AC                        |                            | 81 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.022 AC                        | 37.5 AC                    | 79 AC                   | 0.5 AC               | 0.8793 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.020 AC                        |                            | 76 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.022 AC                        |                            | 78 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.020 AC                        |                            | 75 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.020 AC                        |                            | 69 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

## DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 20       | 16028  | 68308    | 20      | 971                | 3536              | 5      | 7        | ND        | 1023                 | 88895        |
| 06/12/2017 | 36       | 18752  | 66758    | 19      | 1116               | 3622              | 5      | 7        | ND        | 1183                 | 90315        |
| 28/06/2018 | 35       | 13798  | 68160    | 22      | 1144               | 3644              | 5      | 7        | 1         | 1214                 | 86816        |

## ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

## RECOMMENDATION:

RETEST 6 MONTHS: THE DETECTION OF ACETYLENE IS SOME CAUSE FOR CONCERN. CLOSE MONITORING IS NEEDED TO DIAGNOSE THE FAULT

## WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 37     | 12  | 0.78  | A     |
| 30/10/2017 | 29     | 11  | 1.15  | A     |
| 27/11/2017 | 37     | 12  | 0.78  | A     |
| 06/12/2017 | 39     | 12  | 0.66  | A     |
| 22/05/2018 | 29     | 14  | 1.47  | B     |
| 28/06/2018 | 33     | 10  | 0.82  | A     |
| 24/07/2018 | 32     | 11  | 1.04  | A     |
| 25/08/2018 | 29     | 8   | 0.84  | A     |
| 25/09/2018 | 35     | 8   | 0.57  | A     |
| 23/10/2018 | 33     | 8   | 0.66  | A     |

## FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - UAT 3

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |               |
|-------------------|---------------|
| MFG               | - SHANDONG    |
| DATE MFG          | - 2010        |
| MAIN TANK S/N     | - 1035082     |
| KVA               | - 6300        |
| PRI               | - 6300 V      |
| SEC               | - 6300 V      |
| KILOGRAMS         | - 14600       |
| EQUIP TYPE        | - TRANSFORMER |
| TRANS CLASS       | - ONAN        |
| IMPEDENCE         | - 5.78%       |
| PHASE/CYCLE       | - 3/50 Hz     |
| LIQUID TYPE       | - OIL         |
| LITRES LIQUID     | - 3749        |
| LTC MFG           | -             |
| LTC S/N           | -             |
| LTC FAB NO        | -             |
| LTC MODEL         | -             |
| LTC LITRES LIQUID | -             |
| LTC ENERGIZED     | -             |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF  
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | 0.02       | 0.01  | 0.04   |
| 06/12/2017 | ND         | 0.01  | 0.08   |
| 28/06/2018 | ND         | 0.01  | 0.11   |

LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 0.390 AC |
| 06/12/2017 | 0.000 AC | 0.170 AC |
| 28/06/2018 | 0.000 AC | 0.140 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.23 AC    |
| 06/12/2017 | 0.23 AC    |
| 28/06/2018 | 0.21 AC    |

CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE    |
|------------|---------|-------|---------------|
| 04/07/2017 | NON-COR | 3A    | MAGENTA       |
| 06/12/2017 | NON-COR | 2C    | LAVENDER BLUE |
| 28/06/2018 | NON-COR | 3A    | MAGENTA       |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 8.  | TRAFO 1 ESP UNIT 3 | 0073084208    | 11 |

**A) Analysis**

- **Acidity** of 0.020mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 63kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 12ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.14 % M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

**Page Disclaimer :**

*All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein*

DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | ONLINE  |             |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | OFFLINE |             |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 28/11/2017 | ONLINE  |             |            |           |       |        |       |
| 06/12/2017 | ONLINE  |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | OFFLINE |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | OFFLINE |             |            |           |       |        |       |
| 23/10/2018 | ONLINE  |             |            |           |       |        |       |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.025 AC                        |                            | 45 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.025 AC                        |                            | 40 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 28/11/2017 | ROUTINE  | 0.025 AC                        |                            | 31 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.022 AC                        | 34.6 AC                    | 31 AC                   | 1.5 AC               | 0.8180 QU             | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.022 AC                        |                            | 31 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.022 AC                        | 35.0 AC                    | 60 AC                   | 0.5 AC               | 0.8154 QU             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.020 AC                        |                            | 53 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.022 AC                        |                            | 77 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.020 AC                        |                            | 64 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.020 AC                        |                            | 63 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 14       | 12943  | 68807    | 3       | 62                 | 1243              | ND     | ND       | ND        | 79                   | 83072        |
| 06/12/2017 | 14       | 12739  | 65097    | 3       | 66                 | 612               | ND     | ND       | ND        | 83                   | 78531        |
| 28/06/2018 | 19       | 15552  | 64448    | 3       | 57                 | 1391              | ND     | ND       | ND        | 79                   | 81470        |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 34     | 20  | 1.64  | C     |
| 30/10/2017 | 28     | 19  | 2.24  | D     |
| 28/11/2017 | 43     | 34  | 1.53  | C     |
| 06/12/2017 | 43     | 31  | 1.39  | C     |
| 21/05/2018 | 31     | 27  | 2.57  | D     |
| 28/06/2018 | 32     | 10  | 0.95  | B     |
| 24/07/2018 | 26     | 9   | 1.17  | B     |
| 25/08/2018 | 26     | 9   | 1.17  | B     |
| 25/09/2018 | 30     | 10  | 1.05  | B     |
| 23/10/2018 | 31     | 12  | 1.14  | B     |

NAMEPLATE DATA

|                   |                     |
|-------------------|---------------------|
| MFG               | - ZHEJIAN           |
| DATE MFG          | - 2010              |
| MAIN TANK S/N     | - <b>0073084208</b> |
| KVA               | -                   |
| PRI               | - 72000 V           |
| SEC               | - 416 V             |
| KILOGRAMS         | - 1350              |
| EQUIP TYPE        | - TRANSFORMER       |
| TRANS CLASS       | - ONAN              |
| IMPEDENCE         | -                   |
| PHASE/CYCLE       | - 3/50 Hz           |
| LIQUID TYPE       | - OIL               |
| LITRES LIQUID     | - 731               |
| LTC MFG           | -                   |
| LTC S/N           | -                   |
| LTC FAB NO        | -                   |
| LTC MODEL         | -                   |
| LTC LITRES LIQUID | -                   |
| LTC ENERGIZED     | -                   |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - NO  |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |                     |
|-------------------|---------------------|
| MFG               | - ZHEJIAN           |
| DATE MFG          | - 2010              |
| MAIN TANK S/N     | - <b>0073084208</b> |
| KVA               | -                   |
| PRI               | - 72000 V           |
| SEC               | - 416 V             |
| KILOGRAMS         | - 1350              |
| EQUIP TYPE        | - TRANSFORMER       |
| TRANS CLASS       | - ONAN              |
| IMPEDENCE         | -                   |
| PHASE/CYCLE       | - 3/50 Hz           |
| LIQUID TYPE       | - OIL               |
| LITRES LIQUID     | - 731               |
| LTC MFG           | -                   |
| LTC S/N           | -                   |
| LTC FAB NO        | -                   |
| LTC MODEL         | -                   |
| LTC LITRES LIQUID | -                   |
| LTC ENERGIZED     | -                   |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - NO  |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | 0.01       | 0.01  | ND     |
| 06/12/2017 | 0.01       | ND    | ND     |
| 28/06/2018 | ND         | ND    | 0.02   |

LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 0.350 AC |
| 06/12/2017 | 0.000 AC | 0.040 AC |
| 28/06/2018 | 0.000 AC | 0.050 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | ND UN      |
| 06/12/2017 | 0.18 QU    |
| 28/06/2018 | 0.16 QU    |

CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE  |
|------------|---------|-------|-------------|
| 04/07/2017 | NON-COR | 1B    | DARK ORANGE |
| 06/12/2017 | NON-COR | 2E    | BRASSY GOLD |
| 28/06/2018 | NON-COR | 1B    | DARK ORANGE |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 9.  | TRAFO 2 ESP UNIT 3 | 0070084208    | 12 |

**A) Analysis**

- **Acidity** of 0.25mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 54kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 13ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.24% M/dw is graded as **B**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

**Page Disclaimer :**

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DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | ONLINE  |             |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | OFFLINE |             |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 28/11/2017 | ONLINE  |             |            |           |       |        |       |
| 06/12/2017 | ONLINE  |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | OFFLINE |             |            |           |       |        |       |
| 24/07/2018 | ONLINE  |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | ONLINE  |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

NAMEPLATE DATA

|                   |                     |
|-------------------|---------------------|
| MFG               | - ZHEJIAN           |
| DATE MFG          | - 2010              |
| MAIN TANK S/N     | - <b>0070084208</b> |
| KVA               | -                   |
| PRI               | - 72000 V           |
| SEC               | - 415 V             |
| KILOGRAMS         | - 1350              |
| EQUIP TYPE        | - TRANSFORMER       |
| TRANS CLASS       | - ONAN              |
| IMPEDENCE         | -                   |
| PHASE/CYCLE       | - 3/50 Hz           |
| LIQUID TYPE       | - OIL               |
| LITRES LIQUID     | - 731               |
| LTC MFG           | -                   |
| LTC S/N           | -                   |
| LTC FAB NO        | -                   |
| LTC MODEL         | -                   |
| LTC LITRES LIQUID | -                   |
| LTC ENERGIZED     | -                   |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.115 UN                        |                            | 21 UN                   | 2.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.118 UN                        |                            | 33 AC                   | 2.5 AC               |                       | CLEAR AC             |                          |
| 28/11/2017 | ROUTINE  | 0.017 AC                        |                            | 25 QU                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.022 AC                        | 29.7 QU                    | 29 QU                   | 0.5 AC               | 0.8106 QU             | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.022 AC                        |                            | 28 QU                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.025 AC                        | 30.2 QU                    | 34 AC                   | 0.5 AC               | 0.8080 QU             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.022 AC                        |                            | 30 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.025 AC                        |                            | 67 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.022 AC                        |                            | 31 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.025 AC                        |                            | 54 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 38       | 16781  | 70523    | 5       | 129                | 2431              | 3      | 5        | ND        | 180                  | 89915        |
| 06/12/2017 | 42       | 19523  | 66059    | 3       | 102                | 912               | ND     | ND       | ND        | 147                  | 86641        |
| 28/06/2018 | 43       | 22272  | 68719    | 4       | 38                 | 1226              | ND     | ND       | ND        | 85                   | 92302        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 36     | 37  | 2.63  | D     |
| 30/10/2017 | 28     | 20  | 2.36  | D     |
| 28/11/2017 | 43     | 28  | 1.26  | B     |
| 06/12/2017 | 43     | 44  | 1.98  | D     |
| 21/05/2018 | 31     | 30  | 2.85  | E     |
| 28/06/2018 | 30     | 16  | 1.68  | C     |
| 24/07/2018 | 32     | 25  | 2.38  | D     |
| 25/08/2018 | 26     | 11  | 1.43  | C     |
| 25/09/2018 | 39     | 32  | 1.76  | D     |
| 23/10/2018 | 32     | 13  | 1.24  | B     |

FIELD SERVICE

DATE SERVICE  
01/07/2018 PURIFIED

# CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

12

DATE PRINTED 13/11/2018

ID# - 601669

**CUST - PT IP UJP PLTU  
JERANJANG**

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR  
OTHER -

## NAMEPLATE DATA

|                   |                     |
|-------------------|---------------------|
| MFG               | - ZHEJIAN           |
| DATE MFG          | - 2010              |
| MAIN TANK S/N     | - <b>0070084208</b> |
| KVA               | -                   |
| PRI               | - 72000 V           |
| SEC               | - 415 V             |
| KILOGRAMS         | - 1350              |
| EQUIP TYPE        | - TRANSFORMER       |
| TRANS CLASS       | - ONAN              |
| IMPEDENCE         | -                   |
| PHASE/CYCLE       | - 3/50 Hz           |
| LIQUID TYPE       | - OIL               |
| LITRES LIQUID     | - 731               |
| LTC MFG           | -                   |
| LTC S/N           | -                   |
| LTC FAB NO        | -                   |
| LTC MODEL         | -                   |
| LTC LITRES LIQUID | -                   |
| LTC ENERGIZED     | -                   |

## ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

## FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | ND   | 169  | ND   | ND   | 169   |
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

## METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | ND         | 0.01  | 0.01   |
| 06/12/2017 | ND         | ND    | ND     |
| 28/06/2018 | ND         | ND    | 0.02   |

## LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 1.330 AC |
| 06/12/2017 | 0.000 AC | 0.030 AC |
| 28/06/2018 | 0.000 AC | 0.040 AC |

## INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | ND UN      |
| 06/12/2017 | 0.20 AC    |
| 28/06/2018 | 0.19 QU    |

## CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE    |
|------------|---------|-------|---------------|
| 04/07/2017 | NON-COR | 2C    | LAVENDER BLUE |
| 06/12/2017 | NON-COR | 3A    | MAGENTA       |
| 28/06/2018 | NON-COR | 3A    | MAGENTA       |

## PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

## FIELD SERVICE

DATE SERVICE  
01/07/2018 PURIFIED

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 10. | TRAFO 3 ESP UNIT 3 | 0072084208    | 13 |

**A) Analysis**

- **Acidity** of 0.014mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 49kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 15ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.82% M/dw is graded as **A**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates the presence of high water content in the oil.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

**Page Disclaimer :**

*All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein*

DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | ONLINE  |             |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | OFFLINE |             |            | LOW       | FAIR  | GOOD   | NONE  |
| 28/11/2017 | ONLINE  |             |            |           |       |        |       |
| 06/12/2017 | ONLINE  |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | ONLINE  |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | ONLINE  |             |            |           |       |        |       |
| 23/10/2018 | ONLINE  |             |            |           |       |        |       |

NAMEPLATE DATA

|                   |                     |
|-------------------|---------------------|
| MFG               | - ZHEJIAN           |
| DATE MFG          | - 2010              |
| MAIN TANK S/N     | - <b>0072084208</b> |
| KVA               | -                   |
| PRI               | - 72000 V           |
| SEC               | - 415 V             |
| KILOGRAMS         | - 1350              |
| EQUIP TYPE        | - TRANSFORMER       |
| TRANS CLASS       | - ONAN              |
| IMPEDENCE         | -                   |
| PHASE/CYCLE       | - 3/50 Hz           |
| LIQUID TYPE       | - OIL               |
| LITRES LIQUID     | - 731               |
| LTC MFG           | -                   |
| LTC S/N           | -                   |
| LTC FAB NO        | -                   |
| LTC MODEL         | -                   |
| LTC LITRES LIQUID | -                   |
| LTC ENERGIZED     | -                   |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.028 AC                        |                            | 59 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.028 AC                        |                            | 55 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 28/11/2017 | ROUTINE  | 0.014 AC                        |                            | 39 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.014 AC                        | 39.8 AC                    | 45 AC                   | 0.5 AC               | 0.8099 QU             | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.014 AC                        |                            | 54 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.017 AC                        | 38.4 AC                    | 43 AC                   | 0.5 AC               | 0.8075 QU             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.014 AC                        |                            | 43 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.014 AC                        |                            | 76 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.020 AC                        |                            | 47 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.014 AC                        |                            | 49 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 21       | 16364  | 70694    | 3       | 38                 | 1068              | ND     | ND       | ND        | 62                   | 88188        |
| 06/12/2017 | 15       | 14515  | 65277    | 2       | 40                 | 559               | ND     | ND       | ND        | 57                   | 80408        |
| 28/06/2018 | 30       | 15202  | 64188    | 4       | 65                 | 1833              | ND     | ND       | ND        | 99                   | 81322        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 37     | 22  | 1.43  | C     |
| 30/10/2017 | 28     | 16  | 1.89  | D     |
| 28/11/2017 | 44     | 24  | 1.08  | B     |
| 06/12/2017 | 44     | 29  | 1.30  | B     |
| 21/05/2018 | 32     | 23  | 2.18  | D     |
| 28/06/2018 | 35     | 22  | 1.56  | C     |
| 24/07/2018 | 34     | 20  | 1.64  | C     |
| 25/08/2018 | 25     | 11  | 1.43  | C     |
| 25/09/2018 | 40     | 20  | 1.10  | B     |
| 23/10/2018 | 39     | 15  | 0.82  | A     |

FIELD SERVICE

DATE SERVICE  
01/07/2018 PURIFIED

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - TRAFO 3 ESP UNIT 3

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |               |
|-------------------|---------------|
| MFG               | - ZHEJIAN     |
| DATE MFG          | - 2010        |
| MAIN TANK S/N     | - 0072084208  |
| KVA               | -             |
| PRI               | - 72000 V     |
| SEC               | - 415 V       |
| KILOGRAMS         | - 1350        |
| EQUIP TYPE        | - TRANSFORMER |
| TRANS CLASS       | - ONAN        |
| IMPEDENCE         | -             |
| PHASE/CYCLE       | - 3/50 Hz     |
| LIQUID TYPE       | - OIL         |
| LITRES LIQUID     | - 731         |
| LTC MFG           | -             |
| LTC S/N           | -             |
| LTC FAB NO        | -             |
| LTC MODEL         | -             |
| LTC LITRES LIQUID | -             |
| LTC ENERGIZED     | -             |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | 4    | ND   | ND   | ND   | 4     |

RECOMMENDATION:  
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A  
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ALUMINIUM | IRON | COPPER |
|------------|-----------|------|--------|
| 04/07/2017 | ND        | 0.01 | ND     |
| 06/12/2017 | ND        | ND   | ND     |
| 28/06/2018 | ND        | ND   | 0.02   |

LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 0.620 AC |
| 06/12/2017 | 0.000 AC | 0.080 AC |
| 28/06/2018 | 0.000 AC | 0.010 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.01 UN    |
| 06/12/2017 | 0.21 AC    |
| 28/06/2018 | 0.20 AC    |

CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE    |
|------------|---------|-------|---------------|
| 04/07/2017 | NON-COR | 2C    | LAVENDER BLUE |
| 06/12/2017 | NON-COR | 2E    | BRASSY GOLD   |
| 28/06/2018 | NON-COR | 2E    | BRASSY GOLD   |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

FIELD SERVICE

DATE SERVICE  
01/07/2018 PURIFIED

| No. | Unit No.       | Serial Number | TC |
|-----|----------------|---------------|----|
| 11. | TRAFO 4 UNIT 3 | 0071084208    | 14 |

**A) Analysis**

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 56kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water content** in oil of 12ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.26% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.

**B) Recommendation**

- Do carry-out \*Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

**Page Disclaimer :**

*All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein*

DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 29/09/2017 | OFFLINE |             |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 30/10/2017 | OFFLINE |             |            | NORMAL    | FAIR  | GOOD   | NONE  |
| 28/11/2017 | OFFLINE |             |            |           |       |        |       |
| 06/12/2017 | OFFLINE |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | ONLINE  |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.022 AC                        |                            | 59 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.025 AC                        |                            | 55 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 28/11/2017 | ROUTINE  | 0.022 AC                        |                            | 69 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.025 AC                        | 29.8 QU                    | 43 AC                   | 1.0 AC               | 0.8471 AC             | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.028 AC                        |                            | 37 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.025 AC                        | 30.5 QU                    | 71 AC                   | 1.0 AC               | 0.8442 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.022 AC                        |                            | 42 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.025 AC                        |                            | 66 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.022 AC                        |                            | 51 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.022 AC                        |                            | 56 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 10       | 18669  | 66036    | 3       | 24                 | 665               | ND     | ND       | ND        | 37                   | 85407        |
| 06/12/2017 | 14       | 15414  | 63712    | 2       | 22                 | 675               | ND     | ND       | ND        | 38                   | 79839        |
| 28/06/2018 | 20       | 14690  | 67822    | 2       | 22                 | 683               | ND     | ND       | ND        | 44                   | 83239        |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 29     | 15  | 1.58  | C     |
| 30/10/2017 | 28     | 15  | 1.77  | D     |
| 28/11/2017 | 29     | 16  | 1.68  | C     |
| 06/12/2017 | 32     | 19  | 1.80  | D     |
| 21/05/2018 | 30     | 21  | 2.20  | D     |
| 28/06/2018 | 36     | 11  | 0.78  | A     |
| 24/07/2018 | 26     | 11  | 1.43  | C     |
| 25/08/2018 | 27     | 11  | 1.30  | B     |
| 25/09/2018 | 30     | 11  | 1.15  | B     |
| 23/10/2018 | 30     | 12  | 1.26  | B     |

NAMEPLATE DATA

|                   |                     |
|-------------------|---------------------|
| MFG               | - ZHEJIAN           |
| DATE MFG          | - 2010              |
| MAIN TANK S/N     | - <b>0071084208</b> |
| KVA               | -                   |
| PRI               | - 72000 V           |
| SEC               | - 415 V             |
| KILOGRAMS         | - 1350              |
| EQUIP TYPE        | - TRANSFORMER       |
| TRANS CLASS       | - ONAN              |
| IMPEDENCE         | -                   |
| PHASE/CYCLE       | - 3/50 Hz           |
| LIQUID TYPE       | - OIL               |
| LITRES LIQUID     | - 731               |
| LTC MFG           | -                   |
| LTC S/N           | -                   |
| LTC FAB NO        | -                   |
| LTC MODEL         | -                   |
| LTC LITRES LIQUID | -                   |
| LTC ENERGIZED     | -                   |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

FIELD SERVICE

|      |         |
|------|---------|
| DATE | SERVICE |
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |                     |
|-------------------|---------------------|
| MFG               | - ZHEJIAN           |
| DATE MFG          | - 2010              |
| MAIN TANK S/N     | - <b>0071084208</b> |
| KVA               | -                   |
| PRI               | - 72000 V           |
| SEC               | - 415 V             |
| KILOGRAMS         | - 1350              |
| EQUIP TYPE        | - TRANSFORMER       |
| TRANS CLASS       | - ONAN              |
| IMPEDENCE         | -                   |
| PHASE/CYCLE       | - 3/50 Hz           |
| LIQUID TYPE       | - OIL               |
| LITRES LIQUID     | - 731               |
| LTC MFG           | -                   |
| LTC S/N           | -                   |
| LTC FAB NO        | -                   |
| LTC MODEL         | -                   |
| LTC LITRES LIQUID | -                   |
| LTC ENERGIZED     | -                   |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | ND         | ND    | ND     |
| 06/12/2017 | ND         | 0.03  | ND     |
| 28/06/2018 | ND         | 0.03  | 0.02   |

LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 1.360 AC |
| 06/12/2017 | 0.000 AC | 0.370 AC |
| 28/06/2018 | 0.000 AC | 0.550 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.01 UN    |
| 06/12/2017 | 0.03 UN    |
| 28/06/2018 | ND UN      |

CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE  |
|------------|---------|-------|-------------|
| 04/07/2017 | NON-COR | 3A    | MAGENTA     |
| 06/12/2017 | NON-COR | 1B    | DARK ORANGE |
| 28/06/2018 | NON-COR | 1B    | DARK ORANGE |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

| No. | Unit No.       | Serial Number | TC |
|-----|----------------|---------------|----|
| 12. | ASH HANDLING-1 | 10820108      | 15 |

**A) Analysis**

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 28kV is **QUESTIONABLE**. This is in line with the amount of moisture detected in the mineral oil.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 29ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.06% M/dw is graded as **C**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.

**B) Recommendation**

- Retest **Water Content** and **Dielectric** in 6 months to assess if corrective work on the transformer oil is required.
- Other parameters: Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

**Page Disclaimer :**

*All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein*

DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - ASH HANDLING-1

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 30/08/2017 | OFFLINE | 30          |            | NORMAL    | FAIR  |        | NONE  |
| 29/09/2017 | OFFLINE | 40          |            | NORMAL    | FAIR  |        | NONE  |
| 30/10/2017 | ONLINE  | 45          |            | NORMAL    | FAIR  |        | NONE  |
| 27/11/2017 | ONLINE  | 46          |            |           |       |        |       |
| 06/12/2017 | ONLINE  | 52          |            |           |       |        |       |
| 22/05/2018 | ONLINE  | 45          |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | ONLINE  |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 23/10/2018 | ONLINE  |             |            |           |       |        |       |

NAMEPLATE DATA

|                   |               |
|-------------------|---------------|
| MFG               | - WOLONG      |
| DATE MFG          | - 2011        |
| MAIN TANK S/N     | - 10820108    |
| KVA               | - 1600        |
| PRI               | - 6300 V      |
| SEC               | - 400 V       |
| KILOGRAMS         | - 4505        |
| EQUIP TYPE        | - TRANSFORMER |
| TRANS CLASS       | - ONAN        |
| IMPEDENCE         | - 5.56%       |
| PHASE/CYCLE       | - 3/50 Hz     |
| LIQUID TYPE       | - OIL         |
| LITRES LIQUID     | - 1023        |
| LTC MFG           | -             |
| LTC S/N           | -             |
| LTC FAB NO        | -             |
| LTC MODEL         | -             |
| LTC LITRES LIQUID | -             |
| LTC ENERGIZED     | -             |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.022 AC                        |                            | 57 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.022 AC                        |                            | 45 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 27/11/2017 | ROUTINE  | 0.020 AC                        |                            | 36 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.017 AC                        | 37.5 AC                    | 34 AC                   | 1.0 AC               | 0.8832 AC             | CLEAR AC             |                          |
| 22/05/2018 | ROUTINE  | 0.022 AC                        |                            | 33 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.025 AC                        | 37.5 AC                    | 42 AC                   | 1.0 AC               | 0.8804 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.022 AC                        |                            | 39 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.025 AC                        |                            | 60 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.025 AC                        |                            | 43 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.022 AC                        |                            | 28 QU                   | 1.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 35       | 10984  | 71058    | 20      | 441                | 5037              | 7      | 8        | ND        | 511                  | 87590        |
| 06/12/2017 | 49       | 12034  | 67124    | 20      | 558                | 5440              | 8      | 8        | ND        | 643                  | 85241        |
| 28/06/2018 | 53       | 10939  | 69394    | 20      | 557                | 5425              | 9      | 8        | ND        | 647                  | 86405        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

RECOMMENDATION:  
GAS LEVELS HAVE REMAINED STABLE. PROCEED WITH ANNUAL MONITORING.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 34     | 16  | 1.31  | B     |
| 30/10/2017 | 33     | 20  | 1.64  | B     |
| 27/11/2017 | 36     | 29  | 2.06  | C     |
| 06/12/2017 | 36     | 26  | 1.85  | B     |
| 22/05/2018 | 33     | 26  | 2.13  | C     |
| 28/06/2018 | 32     | 24  | 2.28  | C     |
| 24/07/2018 | 32     | 20  | 1.90  | B     |
| 25/08/2018 | 30     | 16  | 1.68  | B     |
| 25/09/2018 | 35     | 17  | 1.21  | A     |
| 23/10/2018 | 36     | 29  | 2.06  | C     |

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **ASH HANDLING-1**

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |                   |
|-------------------|-------------------|
| MFG               | - WOLONG          |
| DATE MFG          | - 2011            |
| MAIN TANK S/N     | - <b>10820108</b> |
| KVA               | - 1600            |
| PRI               | - 6300 V          |
| SEC               | - 400 V           |
| KILOGRAMS         | - 4505            |
| EQUIP TYPE        | - TRANSFORMER     |
| TRANS CLASS       | - ONAN            |
| IMPEDENCE         | - 5.56%           |
| PHASE/CYCLE       | - 3/50 Hz         |
| LIQUID TYPE       | - OIL             |
| LITRES LIQUID     | - 1023            |
| LTC MFG           | -                 |
| LTC S/N           | -                 |
| LTC FAB NO        | -                 |
| LTC MODEL         | -                 |
| LTC LITRES LIQUID | -                 |
| LTC ENERGIZED     | -                 |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | ND         | 0.01  | 0.01   |
| 06/12/2017 | ND         | ND    | 0.01   |
| 28/06/2018 | ND         | 0.01  | 0.04   |

LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 0.730 AC |
| 06/12/2017 | 0.000 AC | 0.240 AC |
| 28/06/2018 | 0.000 AC | 0.460 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.23 AC    |
| 06/12/2017 | 0.24 AC    |
| 28/06/2018 | 0.22 AC    |

CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE    |
|------------|---------|-------|---------------|
| 04/07/2017 | NON-COR | 1B    | DARK ORANGE   |
| 06/12/2017 | NON-COR | 2C    | LAVENDER BLUE |
| 28/06/2018 | NON-COR | 1B    | DARK ORANGE   |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

| No. | Unit No.       | Serial Number | TC |
|-----|----------------|---------------|----|
| 13. | ASH HANDLING-2 | 10820107      | 16 |

**A) Analysis**

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 36kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 23ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.50% M/dw is graded as **B**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

**Page Disclaimer :**

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DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - ASH HANDLING-2

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 30/08/2017 | ONLINE  | 47          |            | NORMAL    | FAIR  |        | NONE  |
| 29/09/2017 | ONLINE  | 60          |            | NORMAL    | FAIR  |        | NONE  |
| 30/10/2017 | OFFLINE | 32          |            | NORMAL    | FAIR  |        | NONE  |
| 27/11/2017 | OFFLINE | 40          |            |           |       |        |       |
| 06/12/2017 | ONLINE  | 48          |            |           |       |        |       |
| 22/05/2018 | ONLINE  | 45          |            |           |       | GOOD   | NONE  |
| 28/06/2018 | OFFLINE |             |            |           |       |        |       |
| 24/07/2018 | ONLINE  |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | OFFLINE |             |            |           |       |        |       |

NAMEPLATE DATA

|                   |                   |
|-------------------|-------------------|
| MFG               | - WOLONG          |
| DATE MFG          | - 2011            |
| MAIN TANK S/N     | - <b>10820107</b> |
| KVA               | - 1600            |
| PRI               | - 6300 V          |
| SEC               | - 400 V           |
| KILOGRAMS         | - 4505            |
| EQUIP TYPE        | - TRANSFORMER     |
| TRANS CLASS       | - ONAN            |
| IMPEDENCE         | - 5.6%            |
| PHASE/CYCLE       | - 3/50 Hz         |
| LIQUID TYPE       | - OIL             |
| LITRES LIQUID     | - 1023            |
| LTC MFG           | -                 |
| LTC S/N           | -                 |
| LTC FAB NO        | -                 |
| LTC MODEL         | -                 |
| LTC LITRES LIQUID | -                 |
| LTC ENERGIZED     | -                 |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 29/09/2017 | ROUTINE  | 0.022 AC                        |                            | 28 QU                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 30/10/2017 | ROUTINE  | 0.020 AC                        |                            | 46 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 27/11/2017 | ROUTINE  | 0.022 AC                        |                            | 47 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.020 AC                        | 36.6 AC                    | 24 UN                   | 1.5 AC               | 0.8840 AC             | CLEAR AC             |                          |
| 22/05/2018 | ROUTINE  | 0.020 AC                        |                            | 24 UN                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.022 AC                        | 35.0 AC                    | 50 AC                   | 1.5 AC               | 0.8811 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.020 AC                        |                            | 27 QU                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.022 AC                        |                            | 45 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.022 AC                        |                            | 36 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.022 AC                        |                            | 36 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 18       | 21157  | 64224    | 4       | 129                | 1676              | ND     | 1        | ND        | 152                  | 87209        |
| 06/12/2017 | 20       | 15791  | 63530    | 5       | 275                | 2783              | ND     | 4        | ND        | 304                  | 82408        |
| 28/06/2018 | 32       | 22904  | 67139    | 4       | 249                | 3112              | ND     | 5        | ND        | 290                  | 93445        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - NO  |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - NO  |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

RECOMMENDATION:  
CARBON DIOXIDE GAS LEVEL IS HIGH.OTHER GASES ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 29/09/2017 | 40     | 31  | 1.71  | B     |
| 30/10/2017 | 29     | 21  | 2.20  | C     |
| 27/11/2017 | 29     | 20  | 2.10  | C     |
| 06/12/2017 | 34     | 32  | 2.62  | D     |
| 22/05/2018 | 33     | 32  | 2.62  | D     |
| 28/06/2018 | 28     | 16  | 1.89  | B     |
| 24/07/2018 | 33     | 30  | 2.46  | C     |
| 25/08/2018 | 28     | 16  | 1.89  | B     |
| 25/09/2018 | 30     | 19  | 2.00  | B     |
| 23/10/2018 | 37     | 23  | 1.50  | B     |

FIELD SERVICE

DATE SERVICE  
01/07/2018 PURIFIED

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **ASH HANDLING-2**

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |                   |
|-------------------|-------------------|
| MFG               | - WOLONG          |
| DATE MFG          | - 2011            |
| MAIN TANK S/N     | - <b>10820107</b> |
| KVA               | - 1600            |
| PRI               | - 6300 V          |
| SEC               | - 400 V           |
| KILOGRAMS         | - 4505            |
| EQUIP TYPE        | - TRANSFORMER     |
| TRANS CLASS       | - ONAN            |
| IMPEDENCE         | - 5.6%            |
| PHASE/CYCLE       | - 3/50 Hz         |
| LIQUID TYPE       | - OIL             |
| LITRES LIQUID     | - 1023            |
| LTC MFG           | -                 |
| LTC S/N           | -                 |
| LTC FAB NO        | -                 |
| LTC MODEL         | -                 |
| LTC LITRES LIQUID | -                 |
| LTC ENERGIZED     | -                 |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - NO  |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - NO  |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 04/07/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 06/12/2017 | ND   | ND   | 3    | ND   | ND   | 3     |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ®ALUMINIUM | ®IRON | COPPER |
|------------|------------|-------|--------|
| 04/07/2017 | ND         | 0.01  | ND     |
| 06/12/2017 | ND         | 0.01  | 0.01   |
| 28/06/2018 | ND         | ND    | 0.02   |

LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 04/07/2017 | 0.000 AC | 1.350 AC |
| 06/12/2017 | 0.000 AC | 0.270 AC |
| 28/06/2018 | 0.000 AC | 0.490 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 04/07/2017 | 0.22 AC    |
| 06/12/2017 | 0.24 AC    |
| 28/06/2018 | 0.20 AC    |

CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE   |
|------------|---------|-------|--------------|
| 04/07/2017 | NON-COR | 3A    | MAGENTA      |
| 06/12/2017 | NON-COR | 1A    | LIGHT ORANGE |
| 28/06/2018 | NON-COR | 3A    | MAGENTA      |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

FIELD SERVICE

DATE SERVICE  
01/07/2018 PURIFIED

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 14. | TRAFO 1 ESP UNIT 2 | 10037         | 17 |

**A) Analysis**

- **Acidity** of 0.028mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 54kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 11ppm is **ACCEPTABLE** while moisture in solid insulation of 1.04% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes.**

**Page Disclaimer :**

*All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein*

DATE PRINTED 12/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 28/11/2017 | OFFLINE |             |            |           |       |        |       |
| 06/12/2017 | OFFLINE |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | OFFLINE |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | OFFLINE |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

NAMEPLATE DATA

|               |                |
|---------------|----------------|
| MFG           | - FUJIAN       |
| DATE MFG      | - 2010         |
| MAIN TANK S/N | - <b>10037</b> |
| KVA           | -              |
| PRI           | - 72000 V      |
| SEC           | - 380 V        |
| KILOGRAMS     | - 1000         |
| EQUIP TYPE    | - TRANSFORMER  |
| TRANS CLASS   | - ONAN         |
| IMPEDENCE     | -              |
| PHASE/CYCLE   | - 3/50 Hz      |
| LIQUID TYPE   | - OIL          |
| LITRES LIQUID | - 630          |

|                   |   |
|-------------------|---|
| LTC MFG           | - |
| LTC S/N           | - |
| LTC FAB NO        | - |
| LTC MODEL         | - |
| LTC LITRES LIQUID | - |
| LTC ENERGIZED     | - |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 28/11/2017 | ROUTINE  | 0.020 AC                        |                            | 48 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.022 AC                        |                            | 22 UN                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.025 AC                        |                            | 24 UN                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.025 AC                        | 32.9 AC                    | 77 AC                   | 1.0 AC               | 0.8693 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.028 AC                        |                            | 69 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.031 AC                        |                            | 74 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.028 AC                        |                            | 74 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.028 AC                        |                            | 54 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 28/06/2018 | 93       | 17225  | 43231    | 5       | 57                 | 775               | ND     | 2        | ND        | 157                  | 61388        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 28/11/2017 | 27     | 21  | 2.48  | D     |
| 06/12/2017 | 27     | 35  | 4.13  | E     |
| 21/05/2018 | 31     | 32  | 3.04  | E     |
| 28/06/2018 | 28     | 11  | 1.30  | B     |
| 24/07/2018 | 31     | 11  | 1.04  | B     |
| 25/08/2018 | 29     | 12  | 1.26  | B     |
| 25/09/2018 | 30     | 10  | 1.05  | B     |
| 23/10/2018 | 31     | 11  | 1.04  | B     |

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 12/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10037</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

|            |      |      |      |      |      |       |
|------------|------|------|------|------|------|-------|
| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

|            |            |       |        |
|------------|------------|-------|--------|
| DATE       | ®ALUMINIUM | ®IRON | COPPER |
| 28/06/2018 | ND         | 0.06  | 0.03   |

LIQUID POWER FACTOR (ASTM D924)

|            |          |          |
|------------|----------|----------|
| DATE       | 25°C     | 100°C    |
| 28/06/2018 | 0.030 AC | 0.420 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

|            |            |
|------------|------------|
| DATE       | %BY WEIGHT |
| 28/06/2018 | ND UN      |

CORROSIVE SULFUR (ASTM D1275)

|            |         |       |               |
|------------|---------|-------|---------------|
| DATE       | STATUS  | CLASS | APPEARANCE    |
| 28/06/2018 | NON-COR | 2C    | LAVENDER BLUE |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

|      |     |
|------|-----|
| DATE | PPM |
|------|-----|

FIELD SERVICE

|      |         |
|------|---------|
| DATE | SERVICE |
|------|---------|

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 15. | TRAFO 2 ESP UNIT 2 | 10036         | 18 |

**A) Analysis**

- **Acidity** of 0.062mgKOH/gm is **QUESTIONABLE**. The levels remained stable without significant changes. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 67kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 13ppm is **ACCEPTABLE** while moisture in solid insulation of 1.24% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

**B) Recommendation**

- Do carry-out \*Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

**Page Disclaimer :**

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DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 28/11/2017 | OFFLINE |             |            |           |       |        |       |
| 06/12/2017 | OFFLINE |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | OFFLINE |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

NAMEPLATE DATA

|               |                |
|---------------|----------------|
| MFG           | - FUJIAN       |
| DATE MFG      | - 2010         |
| MAIN TANK S/N | - <b>10036</b> |
| KVA           | -              |
| PRI           | - 72000 V      |
| SEC           | - 380 V        |
| KILOGRAMS     | - 1000         |
| EQUIP TYPE    | - TRANSFORMER  |
| TRANS CLASS   | - ONAN         |
| IMPEDENCE     | -              |
| PHASE/CYCLE   | - 3/50 Hz      |
| LIQUID TYPE   | - OIL          |
| LITRES LIQUID | - 630          |

|                   |   |
|-------------------|---|
| LTC MFG           | - |
| LTC S/N           | - |
| LTC FAB NO        | - |
| LTC MODEL         | - |
| LTC LITRES LIQUID | - |
| LTC ENERGIZED     | - |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 28/11/2017 | ROUTINE  | 0.048 AC                        |                            | 55 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.051 QU                        |                            | 59 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.053 QU                        |                            | 55 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.056 QU                        | 31.3 QU                    | 66 AC                   | 1.0 AC               | 0.8573 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.059 QU                        |                            | 79 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.062 QU                        |                            | 67 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.056 QU                        |                            | 51 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.062 QU                        |                            | 67 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 28/06/2018 | 8        | 22714  | 64906    | 5       | 67                 | 879               | ND     | ND       | ND        | 80                   | 88579        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 28/11/2017 | 28     | 13  | 1.53  | C     |
| 06/12/2017 | 28     | 14  | 1.65  | C     |
| 21/05/2018 | 31     | 16  | 1.52  | C     |
| 28/06/2018 | 27     | 13  | 1.53  | C     |
| 24/07/2018 | 31     | 8   | 0.76  | A     |
| 25/08/2018 | 31     | 10  | 0.95  | B     |
| 25/09/2018 | 29     | 9   | 0.94  | B     |
| 23/10/2018 | 31     | 13  | 1.24  | B     |

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10036</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

|            |      |      |      |      |      |       |
|------------|------|------|------|------|------|-------|
| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
| 28/06/2018 | ND   | 42   | ND   | ND   | ND   | 42    |

RECOMMENDATION:  
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A  
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

|            |            |       |        |
|------------|------------|-------|--------|
| DATE       | ®ALUMINIUM | ®IRON | COPPER |
| 28/06/2018 | ND         | ND    | 0.02   |

LIQUID POWER FACTOR (ASTM D924)

|            |          |          |
|------------|----------|----------|
| DATE       | 25°C     | 100°C    |
| 28/06/2018 | 0.000 AC | 0.680 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

|            |            |
|------------|------------|
| DATE       | %BY WEIGHT |
| 28/06/2018 | ND UN      |

CORROSIVE SULFUR (ASTM D1275)

|            |         |       |            |
|------------|---------|-------|------------|
| DATE       | STATUS  | CLASS | APPEARANCE |
| 28/06/2018 | NON-COR | 3A    | MAGENTA    |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

|      |     |
|------|-----|
| DATE | PPM |
|------|-----|

FIELD SERVICE

|      |         |
|------|---------|
| DATE | SERVICE |
|------|---------|

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 16. | TRAFO 3 ESP UNIT 2 | 10033         | 19 |

**A) Analysis**

- **Acidity** of 0.070mgKOH/g is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 86kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 13ppm is **ACCEPTABLE** while moisture in solid insulation of 1.24% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

**B) Recommendation**

- Based on previous oil analysis, the IFT of the oil is unfavorable. Together with the high level of Acidity, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

**Page Disclaimer :**

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DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 28/11/2017 | OFFLINE |             |            |           |       |        |       |
| 06/12/2017 | OFFLINE |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 25/09/2018 | OFFLINE |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

NAMEPLATE DATA

|               |                |
|---------------|----------------|
| MFG           | - FUJIAN       |
| DATE MFG      | - 2010         |
| MAIN TANK S/N | - <b>10033</b> |
| KVA           | -              |
| PRI           | - 72000 V      |
| SEC           | - 380 V        |
| KILOGRAMS     | - 1000         |
| EQUIP TYPE    | - TRANSFORMER  |
| TRANS CLASS   | - ONAN         |
| IMPEDENCE     | -              |
| PHASE/CYCLE   | - 3/50 Hz      |
| LIQUID TYPE   | - OIL          |
| LITRES LIQUID | - 630          |

|                   |   |
|-------------------|---|
| LTC MFG           | - |
| LTC S/N           | - |
| LTC FAB NO        | - |
| LTC MODEL         | - |
| LTC LITRES LIQUID | - |
| LTC ENERGIZED     | - |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 28/11/2017 | ROUTINE  | 0.053 QU                        |                            | 42 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.056 QU                        |                            | 60 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.059 QU                        |                            | 51 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.062 QU                        | 26.3 UN                    | 78 AC                   | 1.5 AC               | 0.8602 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.070 QU                        |                            | 67 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.067 QU                        |                            | 73 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.067 QU                        |                            | 59 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.070 QU                        |                            | 86 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 28/06/2018 | 14       | 17627  | 48672    | 4       | 47                 | 935               | ND     | ND       | ND        | 65                   | 67299        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 28/11/2017 | 28     | 15  | 1.77  | D     |
| 06/12/2017 | 28     | 16  | 1.89  | D     |
| 21/05/2018 | 31     | 18  | 1.71  | D     |
| 28/06/2018 | 28     | 14  | 1.65  | C     |
| 24/07/2018 | 32     | 12  | 1.14  | B     |
| 25/08/2018 | 30     | 13  | 1.37  | C     |
| 25/09/2018 | 30     | 12  | 1.26  | B     |
| 23/10/2018 | 32     | 13  | 1.24  | B     |

FIELD SERVICE

|      |         |
|------|---------|
| DATE | SERVICE |
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10033</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

|            |      |      |      |      |      |       |
|------------|------|------|------|------|------|-------|
| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

|            |            |       |        |
|------------|------------|-------|--------|
| DATE       | ®ALUMINIUM | ®IRON | COPPER |
| 28/06/2018 | ND         | ND    | 0.02   |

LIQUID POWER FACTOR (ASTM D924)

|            |          |          |
|------------|----------|----------|
| DATE       | 25°C     | 100°C    |
| 28/06/2018 | 0.000 AC | 0.570 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

|            |            |
|------------|------------|
| DATE       | %BY WEIGHT |
| 28/06/2018 | ND UN      |

CORROSIVE SULFUR (ASTM D1275)

|            |         |       |            |
|------------|---------|-------|------------|
| DATE       | STATUS  | CLASS | APPEARANCE |
| 28/06/2018 | NON-COR | 3A    | MAGENTA    |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

|      |     |
|------|-----|
| DATE | PPM |
|------|-----|

FIELD SERVICE

|      |         |
|------|---------|
| DATE | SERVICE |
|------|---------|

| No. | Unit No.           | Serial Number | TC |
|-----|--------------------|---------------|----|
| 17. | TRAFO 4 ESP UNIT 2 | 10032         | 20 |

**A) Analysis**

- **Acidity** of 0.034mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 57kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 15ppm is **ACCEPTABLE** while moisture in solid insulation of 1.42% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

**Page Disclaimer :**

*All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein*

DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 28/11/2017 | OFFLINE |             |            |           |       |        |       |
| 06/12/2017 | OFFLINE |             |            |           |       |        |       |
| 21/05/2018 | OFFLINE |             |            |           |       | GOOD   | NONE  |
| 28/06/2018 | ONLINE  |             |            |           |       |        |       |
| 24/07/2018 | OFFLINE |             |            |           |       |        |       |
| 25/08/2018 | OFFLINE |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10032</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 28/11/2017 | ROUTINE  | 0.020 AC                        |                            | 62 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.022 AC                        |                            | 58 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 21/05/2018 | ROUTINE  | 0.025 AC                        |                            | 40 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.028 AC                        | 32.6 AC                    | 88 AC                   | 1.0 AC               | 0.8648 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.031 AC                        |                            | 59 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.031 AC                        |                            | 68 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.031 AC                        |                            | 51 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.034 AC                        |                            | 57 AC                   | 1.0 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 28/06/2018 | 22       | 25201  | 74464    | 5       | 85                 | 823               | ND     | ND       | ND        | 112                  | 100600       |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 28/11/2017 | 28     | 16  | 1.89  | D     |
| 06/12/2017 | 28     | 17  | 2.01  | D     |
| 21/05/2018 | 31     | 18  | 1.71  | D     |
| 28/06/2018 | 27     | 15  | 1.77  | D     |
| 24/07/2018 | 32     | 13  | 1.24  | B     |
| 25/08/2018 | 31     | 10  | 0.95  | B     |
| 25/09/2018 | 29     | 11  | 1.15  | B     |
| 23/10/2018 | 32     | 15  | 1.42  | C     |

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10032</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - NO  |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

|            |      |      |      |      |      |       |
|------------|------|------|------|------|------|-------|
| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
| 28/06/2018 | ND   | ND   | ND   | ND   | ND   | ND    |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

|            |            |       |        |
|------------|------------|-------|--------|
| DATE       | ®ALUMINIUM | ®IRON | COPPER |
| 28/06/2018 | ND         | ND    | 0.03   |

LIQUID POWER FACTOR (ASTM D924)

|            |          |          |
|------------|----------|----------|
| DATE       | 25°C     | 100°C    |
| 28/06/2018 | 0.000 AC | 1.620 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

|            |            |
|------------|------------|
| DATE       | %BY WEIGHT |
| 28/06/2018 | 0.01 UN    |

CORROSIVE SULFUR (ASTM D1275)

|            |         |       |               |
|------------|---------|-------|---------------|
| DATE       | STATUS  | CLASS | APPEARANCE    |
| 28/06/2018 | NON-COR | 2C    | LAVENDER BLUE |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

|      |     |
|------|-----|
| DATE | PPM |
|------|-----|

FIELD SERVICE

|      |         |
|------|---------|
| DATE | SERVICE |
|------|---------|

| No. | Unit No.          | Serial Number | TC |
|-----|-------------------|---------------|----|
| 18. | MAIN TRAFO UNIT 2 | 1006902       | 21 |

**A) Analysis**

- **Acidity** of 0.011mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 64kV is **ACCEPTABLE**.
- **Colour** and **Visualization** of oil is in the **ACCEPTABLE** range.
- **Water Content** in oil of 7ppm is **ACCEPTABLE** while moisture in solid insulation of 0.74% M/dw is graded as **A**. For this transformer, it is recommended to maintain the water content less than 20ppm.

**B) Recommendation**

- Do carry out \*Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

**Page Disclaimer :**

*All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein*

DATE PRINTED 13/11/2018  
ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - MAIN TRAFU UNIT 2

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 27/11/2017 | ONLINE  | 50          | 55         |           |       |        |       |
| 06/12/2017 | ONLINE  |             |            |           |       |        |       |
| 22/05/2018 | ONLINE  |             |            | NORMAL    |       | GOOD   | NONE  |
| 28/06/2018 | OFFLINE | 50          | 45         |           |       |        |       |
| 24/07/2018 | ONLINE  |             |            |           |       | GOOD   |       |
| 25/08/2018 | ONLINE  |             |            |           |       |        |       |
| 25/09/2018 | ONLINE  |             |            |           |       |        |       |
| 23/10/2018 | OFFLINE |             |            |           |       |        |       |

NAMEPLATE DATA

|               |               |
|---------------|---------------|
| MFG           | - WOLONG      |
| DATE MFG      | - 2011        |
| MAIN TANK S/N | - 1006902     |
| KVA           | - 33000       |
| PRI           | - 150000 V    |
| SEC           | - 6300 V      |
| KILOGRAMS     | - 81000       |
| EQUIP TYPE    | - TRANSFORMER |
| TRANS CLASS   | - OFAF        |
| IMPEDENCE     | - 13.03%      |
| PHASE/CYCLE   | - 3/50 Hz     |
| LIQUID TYPE   | - OIL         |
| LITRES LIQUID | - 30857       |

|                   |   |
|-------------------|---|
| LTC MFG           | - |
| LTC S/N           | - |
| LTC FAB NO        | - |
| LTC MODEL         | - |
| LTC LITRES LIQUID | - |
| LTC ENERGIZED     | - |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 27/11/2017 | ROUTINE  | 0.008 AC                        |                            | 70 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 06/12/2017 | ROUTINE  | 0.008 AC                        |                            | 67 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 22/05/2018 | ROUTINE  | 0.011 AC                        |                            | 46 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 28/06/2018 | ROUTINE  | 0.008 AC                        | 42.1 AC                    | 75 AC                   | 0.5 AC               | 0.8663 AC             | CLEAR AC             |                          |
| 24/07/2018 | ROUTINE  | 0.011 AC                        |                            | 79 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/08/2018 | ROUTINE  | 0.011 AC                        |                            | 83 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 25/09/2018 | ROUTINE  | 0.011 AC                        |                            | 90 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |
| 23/10/2018 | ROUTINE  | 0.011 AC                        |                            | 64 AC                   | 0.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 28/06/2018 | 19       | 12733  | 53386    | 10      | 193                | 2448              | ND     | 2        | ND        | 224                  | 68791        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - YES |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - YES |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - YES |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 27/11/2017 | 37     | 16  | 1.04  | B     |
| 06/12/2017 | 38     | 13  | 0.85  | A     |
| 22/05/2018 | 33     | 14  | 1.15  | B     |
| 28/06/2018 | 28     | 10  | 1.18  | B     |
| 24/07/2018 | 33     | 11  | 0.90  | B     |
| 25/08/2018 | 34     | 9   | 0.74  | A     |
| 25/09/2018 | 32     | 7   | 0.67  | A     |
| 23/10/2018 | 29     | 7   | 0.74  | A     |

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 13/11/2018

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - MAIN TRAF0 UNIT 2

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |               |
|-------------------|---------------|
| MFG               | - WOLONG      |
| DATE MFG          | - 2011        |
| MAIN TANK S/N     | - 1006902     |
| KVA               | - 33000       |
| PRI               | - 150000 V    |
| SEC               | - 6300 V      |
| KILOGRAMS         | - 81000       |
| EQUIP TYPE        | - TRANSFORMER |
| TRANS CLASS       | - OFAF        |
| IMPEDENCE         | - 13.03%      |
| PHASE/CYCLE       | - 3/50 Hz     |
| LIQUID TYPE       | - OIL         |
| LITRES LIQUID     | - 30857       |
| LTC MFG           | -             |
| LTC S/N           | -             |
| LTC FAB NO        | -             |
| LTC MODEL         | -             |
| LTC LITRES LIQUID | -             |
| LTC ENERGIZED     | -             |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - YES |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - YES |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - YES |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - YES |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|------------|------|------|------|------|------|-------|
| 06/12/2017 | ND   | ND   | ND   | ND   | ND   | ND    |
| 28/06/2018 | ND   | ND   | ND   | ND   | 113  | 113   |

RECOMMENDATION:  
RETEST 6 MONTHS: THE PRESENCE OF 5M2F INDICATES THE POSSIBILITY OF HOTSPOT  
OVERHEATING.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

| DATE       | ALUMINIUM | IRON | COPPER |
|------------|-----------|------|--------|
| 28/06/2018 | ND        | ND   | 0.03   |

LIQUID POWER FACTOR (ASTM D924)

| DATE       | 25°C     | 100°C    |
|------------|----------|----------|
| 28/06/2018 | 0.000 AC | 0.410 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

| DATE       | %BY WEIGHT |
|------------|------------|
| 28/06/2018 | 0.22 AC    |

CORROSIVE SULFUR (ASTM D1275)

| DATE       | STATUS  | CLASS | APPEARANCE  |
|------------|---------|-------|-------------|
| 28/06/2018 | NON-COR | 1B    | DARK ORANGE |

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

| DATE | PPM |
|------|-----|
|------|-----|

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

**3.0 Summary of Our Comments & Recommendations:**

| No. | Sub Name           | Unit No.   | TC | Recommendation                                                                     |
|-----|--------------------|------------|----|------------------------------------------------------------------------------------|
| 1.  | MAIN TRAF0 UNIT 1  | 100691     | 4  | • Periodic condition monitoring                                                    |
| 2.  | TRAF0 1 ESP UNIT 1 | 10937      | 5  | • Periodic condition monitoring                                                    |
| 3.  | TRAF0 2 ESP UNIT 1 | 10035      | 6  | • Periodic condition monitoring                                                    |
| 4.  | TRAF0 3 ESP UNIT 1 | 10031      | 7  | • Perform <b>OIL REGENERATION</b>                                                  |
| 5.  | TRAF0 4 ESP UNIT 1 | 10034      | 8  | • Perform <b>OIL REGENERATION</b>                                                  |
| 6.  | MAIN TRAF0 UNIT 3  | 1122170    | 9  | • Periodic condition monitoring                                                    |
| 7.  | UAT 3              | 1035082    | 10 | • Periodic condition monitoring                                                    |
| 8.  | TRAF0 1 ESP UNIT 3 | 0073084208 | 11 | • Periodic condition monitoring                                                    |
| 9.  | TRAF0 2 ESP UNIT 3 | 0070084208 | 12 | • Periodic condition monitoring                                                    |
| 10. | TRAF0 3 ESP UNIT 3 | 0072084208 | 13 | • Periodic condition monitoring                                                    |
| 11. | TRAF0 4 UNIT 3     | 0071084208 | 14 | • Periodic condition monitoring                                                    |
| 12. | ASH HANDLING-1     | 10820108   | 15 | • 6 months: <b>Dielectric and Water Content</b><br>• Periodic condition monitoring |
| 13. | ASH HANDLING-2     | 10820107   | 16 | • Periodic condition monitoring                                                    |
| 14. | TRAF0 1 ESP UNIT 2 | 10037      | 17 | • Periodic condition monitoring                                                    |
| 15. | TRAF0 2 ESP UNIT 2 | 10036      | 18 | • Periodic condition monitoring                                                    |
| 16. | TRAF0 3 ESP UNIT 2 | 10033      | 19 | • Perform <b>OIL REGENERATION</b>                                                  |

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

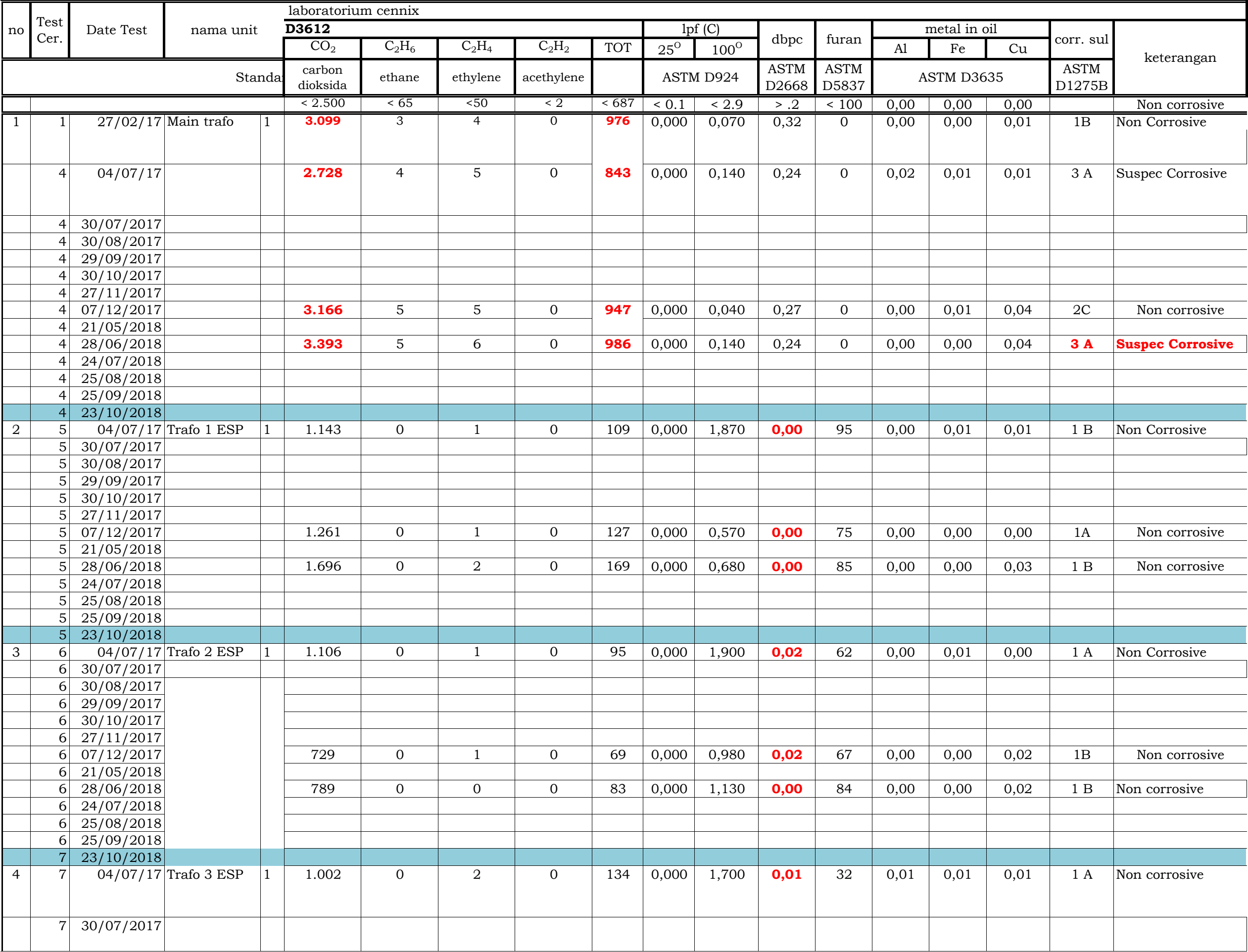
| No. | Sub Name           | Unit No. | TC | Recommendation                  |
|-----|--------------------|----------|----|---------------------------------|
| 17. | TRAFO 4 ESP UNIT 2 | 10032    | 20 | • Periodic condition monitoring |
| 18. | MAIN TRAFO UNIT 2  | 1006902  | 21 | • Periodic condition monitoring |

**Page Disclaimer :**

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| no                    | Test Cer. | Date Test  | nama unit   | Serial number | thn opr | Jumlah minyak, liter | Buatan          | hasil test |           |         |            |             |            |            |            |          |                |                |                |                  |       |     |
|-----------------------|-----------|------------|-------------|---------------|---------|----------------------|-----------------|------------|-----------|---------|------------|-------------|------------|------------|------------|----------|----------------|----------------|----------------|------------------|-------|-----|
|                       |           |            |             |               |         |                      |                 | acid       | ift       | diel.   | color      | spec. Graf. | visual     | sedi.      | moisture   |          | dga ASTM       |                |                |                  |       |     |
|                       |           |            |             |               |         |                      |                 |            |           |         |            |             |            |            | ppm        | grade    | H <sub>2</sub> | O <sub>2</sub> | N <sub>2</sub> | CH <sub>4</sub>  | CO    |     |
| Standard yang dipakai |           |            |             |               |         |                      |                 | ASTM D974  | ASTM D971 | IEC 156 | ASTM D1500 | ASTM D4052  | ASTM D1524 | ASTM D1698 | ASTM D1533 | hydrogen | oksigen        | nitrogen       | methane        | carbon monoksida |       |     |
|                       |           |            |             |               |         |                      | BATASAN OPERASI | < 0,06     | > 32      | > 30    | < 3,5      | > 0,84      | C          |            | < 20       |          | < 100          |                |                | < 120            | < 350 |     |
| 1                     | 1         | 27/02/17   | Main trafo  | 1             | 1006901 | 2011                 | 30.857          | Wolong     | 0,014     | 39,9    | 99         | 0,5         | 0,8693     | C          |            | 9        | A              | 135            | 15.834         | 57.625           | 27    | 807 |
|                       | 4         | 04/07/17   |             |               |         |                      |                 |            | 0,017     | 41,5    | 58         | 0,5         | 0,8691     | C          |            | 10       | A              | 147            | 9.471          | 4.328            | 32    | 655 |
|                       | 4         | 30/07/2017 |             |               |         |                      |                 |            | 0,017     |         | 70         | 0,5         |            | C          |            | 8        | B              |                |                |                  |       |     |
|                       | 4         | 30/08/2017 |             |               |         |                      |                 |            | 0,017     |         | 92         | 0,5         |            | C          |            | 6        | A              |                |                |                  |       |     |
|                       | 4         | 29/09/2017 |             |               |         |                      |                 |            | 0,014     |         | 82         | 0,5         |            | C          |            | 8        | A              |                |                |                  |       |     |
|                       | 4         | 30/10/2017 |             |               |         |                      |                 |            | 0,014     |         | 82         | 0,5         |            | C          |            | 6        | A              |                |                |                  |       |     |
|                       | 4         | 27/11/2017 |             |               |         |                      |                 |            | 0,014     |         | 53         | 0,5         |            | C          |            | 8        | A              |                |                |                  |       |     |
|                       | 4         | 07/12/2017 |             |               |         |                      |                 |            | 0,011     | 40,1    | 87         | 0,5         | 0,8745     | C          |            | 7        | A              | 183            | 12.895         | 43.794           | 39    | 715 |
|                       | 4         | 21/05/2018 |             |               |         |                      |                 |            | 0,011     |         | 77         | 0,5         |            | C          |            | 10       | B              |                |                |                  |       |     |
|                       | 4         | 28/06/2018 |             |               |         |                      |                 |            | 0,014     | 41,9    | 69         | 0,5         | 0,8716     | C          |            | 7        | A              | 176            | 11.044         | 46.776           | 41    | 758 |
|                       | 4         | 24/07/2018 |             |               |         |                      |                 |            | 0,014     |         | 86         | 0,5         |            | C          |            | 6        | A              |                |                |                  |       |     |
|                       | 4         | 25/08/2018 |             |               |         |                      |                 |            | 0,017     |         | 90         | 0,5         |            | C          |            | 6        | A              |                |                |                  |       |     |
|                       | 4         | 25/09/2018 |             |               |         |                      |                 |            | 0,014     |         | 96         | 0,5         |            | C          |            | 7        | A              |                |                |                  |       |     |
|                       | 4         | 23/10/2018 |             |               |         |                      |                 |            | 0,014     |         | 83         | 0,5         |            | C          |            | 8        | A              |                |                |                  |       |     |
| 2                     | 5         | 04/07/17   | Trafo 1 ESP | 1             | 10937   | 2010                 | 1.143           | Fujia      | 0,056     | 25,3    | 50         | 1,5         | 0,8529     | C          |            | 18       | D              | 35             | 20.494         | 67.487           | 4     | 69  |
|                       | 5         | 30/07/2017 |             |               |         |                      |                 |            | 0,059     |         | 57         | 1,5         |            | C          |            | 20       | D              |                |                |                  |       |     |
|                       | 5         | 30/08/2017 |             |               |         |                      |                 |            | 0,056     |         | 65         | 1,5         |            | C          |            | 18       | D              |                |                |                  |       |     |
|                       | 5         | 29/09/2017 |             |               |         |                      |                 |            | 0,059     |         | 58         | 1,5         |            | C          |            | 19       | D              |                |                |                  |       |     |
|                       | 5         | 30/10/2017 |             |               |         |                      |                 |            | 0,059     |         | 23         | 1,5         |            | C          |            | 32       | C              |                |                |                  |       |     |
|                       | 5         | 27/11/2017 |             |               |         |                      |                 |            | 0,062     |         | 25         | 1,5         |            | C          |            | 33       | C              |                |                |                  |       |     |
|                       | 5         | 07/12/2017 |             |               |         |                      |                 |            | 0,062     | 25,0    | 49         | 1,5         | 0,8593     | C          |            | 18       | D              | 39             | 19.933         | 67.198           | 4     | 83  |
|                       | 5         | 21/05/2018 |             |               |         |                      |                 |            | 0,065     |         | 38         | 1,5         |            | C          |            | 33       | E              |                |                |                  |       |     |
|                       | 5         | 28/06/2018 |             |               |         |                      |                 |            | 0,067     | 25,9    | 32         | 1,5         | 0,8560     | C          |            | 30       | E              | 44             | 20.039         | 64.529           | 4     | 119 |
|                       | 5         | 24/07/2018 |             |               |         |                      |                 |            | 0,073     |         | 38         | 1,5         |            | C          |            | 19       | D              |                |                |                  |       |     |
|                       | 5         | 25/08/2018 |             |               |         |                      |                 |            | 0,020     |         | 71         | 0,5         |            | C          |            | 12       | B              |                |                |                  |       |     |
|                       | 5         | 25/09/2018 |             |               |         |                      |                 |            | 0,017     |         | 63         | 0,5         |            | C          |            | 18       | B              |                |                |                  |       |     |
|                       | 5         | 23/10/2018 |             |               |         |                      |                 |            | 0,020     |         | 49         | 0,5         |            | C          |            | 24       | B              |                |                |                  |       |     |
| 3                     | 6         | 04/07/17   | Trafo 2 ESP | 1             | 10035   | 2010                 | 1.143           | Fujia      | 0,042     | 26,6    | 41         | 1,5         | 0,8540     | C          |            | 21       | D              | 27             | 18.639         | 66.781           | 3     | 64  |
|                       | 6         | 30/07/2017 |             |               |         |                      |                 |            | 0,045     |         | 25         | 1,5         |            | C          |            | 27       | E              |                |                |                  |       |     |
|                       | 6         | 30/08/2017 |             |               |         |                      |                 |            | 0,042     |         | 42         | 1,5         |            | C          |            | 14       | C              |                |                |                  |       |     |
|                       | 6         | 29/09/2017 |             |               |         |                      |                 |            | 0,048     |         | 59         | 1,5         |            | C          |            | 19       | D              |                |                |                  |       |     |
|                       | 6         | 30/10/2017 |             |               |         |                      |                 |            | 0,051     |         | 62         | 1,5         |            | C          |            | 17       | B              |                |                |                  |       |     |
|                       | 6         | 27/11/2017 |             |               |         |                      |                 |            | 0,051     |         | 50         | 1,5         |            | C          |            | 18       | B              |                |                |                  |       |     |
|                       | 6         | 07/12/2017 |             |               |         |                      |                 |            | 0,063     | 27,3    | 51         | 1,5         | 0,8589     | C          |            | 17       | D              | 20             | 19.360         | 61.590           | 3     | 45  |
|                       | 6         | 21/05/2018 |             |               |         |                      |                 |            | 0,056     |         | 45         | 1,5         |            | C          |            | 20       | D              |                |                |                  |       |     |
|                       | 6         | 28/06/2018 |             |               |         |                      |                 |            | 0,059     | 27,6    | 55         | 1,5         | 0,8570     | C          |            | 17       | C              | 22             | 22.494         | 64.443           | 4     | 57  |
|                       | 6         | 24/07/2018 |             |               |         |                      |                 |            | 0,062     |         | 47         | 1,5         |            | C          |            | 15       | C              |                |                |                  |       |     |
|                       | 6         | 25/08/2018 |             |               |         |                      |                 |            | 0,022     |         | 56         | 0,5         |            | C          |            | 13       | B              |                |                |                  |       |     |
|                       | 6         | 25/09/2018 |             |               |         |                      |                 |            | 0,020     |         | 67         | 0,5         |            | C          |            | 11       | B              |                |                |                  |       |     |
|                       | 7         | 23/10/2018 |             |               |         |                      |                 |            | 0,020     |         | 71         | 0,5         |            | C          |            | 14       | B              |                |                |                  |       |     |
| 4                     | 7         | 04/07/17   | Trafo 3 ESP | 1             | 10031   | 2010                 | 1.143           | Fujia      | 0,048     | 28,4    | 44         | 1,0         | 0,8537     | C          |            | 32       | E              | 32             | 15.273         | 65.111           | 5     | 95  |
|                       | 7         | 30/07/2017 |             |               |         |                      |                 |            | 0,042     |         | 77         | 1,5         |            |            |            | 10       | D              |                |                |                  |       |     |

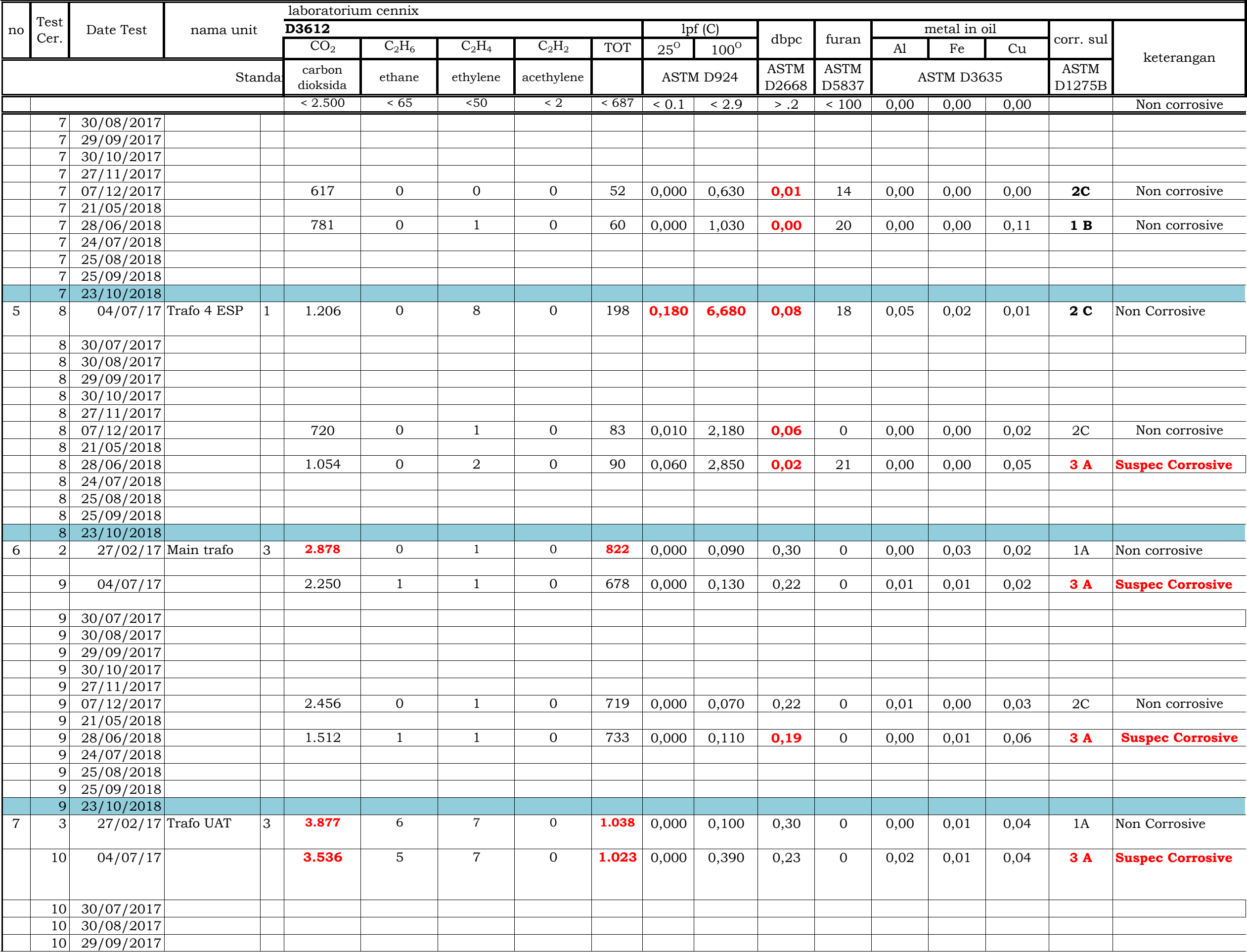




| no        | Test Cer. | Date Test  | nama unit   | MALAYSIAN LAB RECOMMENDATION |                                                                                                                                                        | PAK SOETJAHJONO (TXM ADVISOR)<br>RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------|-----------|------------|-------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standards |           |            |             |                              |                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1         | 1         | 27/02/17   | Main trafo  | 1                            | Retest 6 months: DGA<br>Check leakages<br>Other parameters: Periodic Condition Monitoring                                                              | <p><b>For TEST 23 Oktober 2018</b></p> <p>Keadaan masih sama seperti bulan lalu dimana <b>hampir semua trafo dalam keadaan prima.</b></p> <p>Yang masih perlu mendapat perhatian adalah trafo <b>ESP 3 Unit 1, ESP 2 dan 3 pada Unit 2</b> untuk dilakukan <b>GANTI MINYAK</b>, serta <b>ESP 4 Unit 1</b> untuk di <b>PURIFIKASI</b></p> <p><b>For TEST 25 September 2018</b></p> <p><b>PURIFIKASI</b></p> <p>Pada trafo <b>ESP 4 Unit 1</b> dan <b>ESP 3 Unit 3</b> kemungkinan pekerjaan Purifikasi/Ganti Minyak dilakukan kurang sempurna sehingga terjadi penurunan BDV dan Moisture dengan cepat. Kami sarankan untuk melakukan Purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg diatas 7 cycle.</p> <p><b>GANTI MINYAK</b></p> <p>Kami sarankan untuk trafo <b>ESP 3 Unit 1, ESP 2 Unit 3 dan ESP 2 dan 3 pada Unit 2</b> dilakukan penggantian minyak. seperti pada trafo ESP 2 unti 3, meskipun telah dilakukan Purifikasi, ternyata terjadi penurunan BDV dan Moisture dengan relatif cepat. Kami sarankan pada saat pengisian kembali dilakukan dengan cara purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg diatas 7 cycle.</p> <p><b>For TEST 25 Agustus 2018</b></p> <p>Traf di <b>Unit 1 dan 3</b> pada saat ini dalam <b>kondisi yang optimal</b> untuk operasional setelah dilakukan Purifikasi dan Penggantian Oli. Untuk Trafo <b>ESP 2 dan 3</b> pada <b>Unit 2</b>, kami sarankan untuk dilakukan ganti oli mengingat trend penurunan pada nilai Acidity.</p> <p><b>For TEST 24 Juli 2018</b></p> <p><b>GANTI MINYAK</b></p> <p>Disarankan dilakukan untuk beberapa trafo <b>ESP (Unit 1 no 1 dan 2 , Unit 2 no 3)</b> karena nilai Acid sudah melewati standard. Sedang untuk <b>ESP Unit 2 no 2</b> ada trend Acid yang memburuk, dipersiapkan untuk penggantian oli juga.</p> <p><b>PURIFIKASI</b></p> <p>Disarankan dilakukan pada trafo <b>Ash Handling no 2 , Trafo ESP Unit 1 no 3 dan 4</b>, karena Water Content Grade di bawah standard. Purifikasi agar dilakukan dengan temperatur tinggi dan lebih dari 5 cycle, agar berpengaruh ke Water Content Grade</p> |
|           | 4         | 04/07/17   |             |                              | Retest 6 months: DGA<br>Add inhibitor 2,6-DBPC and Passivator<br>Other parameters: Periodic Condition Monitoring                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 30/07/2017 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 30/08/2017 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 29/09/2017 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 30/10/2017 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 27/11/2017 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 07/12/2017 |             |                              | Retest 6 mnth : DGA                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 21/05/2018 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 28/06/2018 |             |                              | Add passivator                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 24/07/2018 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 25/08/2018 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 25/09/2018 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 4         | 23/10/2018 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2         | 5         | 04/07/17   | Trafo 1 ESP | 1                            | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 30/07/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 30/08/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 29/09/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 30/10/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 27/11/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 07/12/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 21/05/2018 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 28/06/2018 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 24/07/2018 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 25/08/2018 |             |                              | Periodic condition monitoring (change oil on beginning of August 2018)                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 25/09/2018 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 5         | 23/10/2018 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3         | 6         | 04/07/17   | Trafo 2 ESP | 1                            | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 30/07/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 30/08/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 29/09/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 30/10/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 27/11/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 07/12/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 21/05/2018 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 28/06/2018 |             |                              | Perform <b>OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 24/07/2018 |             |                              | <b>Perform OIL REGENERATION</b>                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 25/08/2018 |             |                              | Periodic condition monitoring (change oil on beginning of August 2018)                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 6         | 25/09/2018 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 7         | 23/10/2018 |             |                              | Periodic condition monitoring                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4         | 7         | 04/07/17   | Trafo 3 ESP | 1                            | Retest 6 months: Acidity, IFT, LPF, Diel and Water Content<br>Add inhibitor 2,6-DBPC and Passivator<br>Other parameters: Periodic Condition Monitoring |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|           | 7         | 30/07/2017 |             |                              | Retest 6 months: <b>Acidity, IFT</b> and <b>Liquid Power Factor</b><br>Other parameters: Periodic Condition Monitoring                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



| no                    | Test Cer. | Date Test  | nama unit       | Serial number | thn opr | Jumlah minyak, liter | Buatan | hasil test |           |         |            |             |            |            |            |          |                |                |                |                  |       |     |
|-----------------------|-----------|------------|-----------------|---------------|---------|----------------------|--------|------------|-----------|---------|------------|-------------|------------|------------|------------|----------|----------------|----------------|----------------|------------------|-------|-----|
|                       |           |            |                 |               |         |                      |        | acid       | ift       | diel.   | color      | spec. Graf. | visual     | sedi.      | moisture   |          | dga ASTM       |                |                |                  |       |     |
|                       |           |            |                 |               |         |                      |        |            |           |         |            |             |            |            | ppm        | grade    | H <sub>2</sub> | O <sub>2</sub> | N <sub>2</sub> | CH <sub>4</sub>  | CO    |     |
| Standard yang dipakai |           |            |                 |               |         |                      |        | ASTM D974  | ASTM D971 | IEC 156 | ASTM D1500 | ASTM D4052  | ASTM D1524 | ASTM D1698 | ASTM D1533 | hydrogen | oksigen        | nitrogen       | methane        | carbon monoksida |       |     |
|                       |           |            | BATASAN OPERASI |               |         |                      |        | < 0,06     | > 32      | > 30    | < 3,5      | > 0,84      | C          |            | < 20       |          | < 100          |                |                | < 120            | < 350 |     |
|                       | 7         | 30/08/2017 |                 |               |         |                      |        | 0,042      |           | 61      | 1,5        |             | C          |            | 13         | C        |                |                |                |                  |       |     |
|                       | 7         | 29/09/2017 |                 |               |         |                      |        | 0,045      |           | 72      | 1,5        |             | C          |            | 14         | B        |                |                |                |                  |       |     |
|                       | 7         | 30/10/2017 |                 |               |         |                      |        | 0,048      |           | 72      | 1,5        |             | C          |            | 16         | B        |                |                |                |                  |       |     |
|                       | 7         | 27/11/2017 |                 |               |         |                      |        | 0,051      |           | 62      | 1,5        |             | C          |            | 17         | C        |                |                |                |                  |       |     |
|                       | 7         | 07/12/2017 |                 |               |         |                      |        | 0,063      | 29,3      | 57      | 1,5        | 0,8624      | C          |            | 16         | C        | 7              | 14.730         | 65.738         | 3                | 42    |     |
|                       | 7         | 21/05/2018 |                 |               |         |                      |        | 0,056      |           | 48      | 1,5        |             | C          |            | 21         | D        |                |                |                |                  |       |     |
|                       | 7         | 28/06/2018 |                 |               |         |                      |        | 0,053      | 29,7      | 62      | 1,5        | 0,8605      | C          |            | 17         | D        | 9              | 17.958         | 66.007         | 4                | 46    |     |
|                       | 7         | 24/07/2018 |                 |               |         |                      |        | 0,510      |           | 64      | 1,5        |             | C          |            | 17         | D        |                |                |                |                  |       |     |
|                       | 7         | 25/08/2018 |                 |               |         |                      |        | 0,056      |           | 57      | 1,5        |             | C          |            | 15         | C        |                |                |                |                  |       |     |
|                       | 7         | 25/09/2018 |                 |               |         |                      |        | 0,051      |           | 46      | 1,5        |             | C          |            | 21         | D        |                |                |                |                  |       |     |
|                       | 7         | 23/10/2018 |                 |               |         |                      |        | 0,056      |           | 44      | 1,5        |             | C          |            | 20         | C        |                |                |                |                  |       |     |
| 5                     | 8         | 04/07/17   | Trafo 4 ESP     | 1             | 10034   | 2010                 | 1.143  | Fujia      | 0,028     | 26,3    | 24         | 1,5         | 0,8596     | C          |            | 29       | E              | 13             | 12.265         | 66.770           | 5     | 172 |
|                       | 8         | 30/07/2017 |                 |               |         |                      |        | 0,022      |           | 88      | 1,5        |             | C          |            | 15         | D        |                |                |                |                  |       |     |
|                       | 8         | 30/08/2017 |                 |               |         |                      |        | 0,028      |           | 44      | 1,5        |             | C          |            | 13         | C        |                |                |                |                  |       |     |
|                       | 8         | 29/09/2017 |                 |               |         |                      |        | 0,031      |           | 53      | 1,0        |             | C          |            | 15         | B        |                |                |                |                  |       |     |
|                       | 8         | 30/10/2017 |                 |               |         |                      |        | 0,028      |           | 48      | 1,0        |             | C          |            | 18         | B        |                |                |                |                  |       |     |
|                       | 8         | 27/11/2017 |                 |               |         |                      |        | 0,034      |           | 41      | 1,5        |             | C          |            | 30         | D        |                |                |                |                  |       |     |
|                       | 8         | 07/12/2017 |                 |               |         |                      |        | 0,031      | 27,2      | 41      | 1,5        | 0,8645      | C          |            | 18         | D        | 9              | 13.331         | 63.527         | 3                | 70    |     |
|                       | 8         | 21/05/2018 |                 |               |         |                      |        | 0,034      |           | 37      | 1,5        |             | C          |            | 23         | D        |                |                |                |                  |       |     |
|                       | 8         | 28/06/2018 |                 |               |         |                      |        | 0,037      | 27,5      | 55      | 1,5        | 0,8653      | C          |            | 20         | D        | 9              | 17.962         | 64.248         | 4                | 75    |     |
|                       | 8         | 24/07/2018 |                 |               |         |                      |        | 0,039      |           | 50      | 1,5        |             | C          |            | 19         | D        |                |                |                |                  |       |     |
|                       | 8         | 25/08/2018 |                 |               |         |                      |        | 0,017      |           | 61      | 0,5        |             | C          |            | 14         | C        |                |                |                |                  |       |     |
|                       | 8         | 25/09/2018 |                 |               |         |                      |        | 0,014      |           | 45      | 0,5        |             | C          |            | 20         | D        |                |                |                |                  |       |     |
|                       | 8         | 23/10/2018 |                 |               |         |                      |        | 0,014      |           | 52      | 0,5        |             | C          |            | 21         | B        |                |                |                |                  |       |     |
| 6                     | 2         | 27/02/17   | Main trafo      | 3             | 1122170 | 2010                 | 26.706 | Shandong   | 0,017     | 39,7    | 87         | 0,5         | 0,8781     | C          |            | 9        | A              | 4              | 13.330         | 43.140           | 9     | 808 |
|                       | 9         | 04/07/17   |                 |               |         |                      |        | 0,017      | 42,3      | 78      | 0,5        | 0,8780      | C          |            | 15         | B        | 24             | 10.131         | 44.084         | 11               | 641   |     |
|                       | 9         | 30/07/2017 |                 |               |         |                      |        | 0,014      |           | 77      | 0,5        |             | C          |            | 17         | B        |                |                |                |                  |       |     |
|                       | 9         | 30/08/2017 |                 |               |         |                      |        | 0,017      |           | 86      | 0,5        |             | C          |            | 8          | A        |                |                |                |                  |       |     |
|                       | 9         | 29/09/2017 |                 |               |         |                      |        | 0,014      |           | 86      | 0,5        |             | C          |            | 10         | A        |                |                |                |                  |       |     |
|                       | 9         | 30/10/2017 |                 |               |         |                      |        | 0,014      |           | 88      | 0,5        |             | C          |            | 9          | A        |                |                |                |                  |       |     |
|                       | 9         | 27/11/2017 |                 |               |         |                      |        | 0,014      |           | 87      | 0,5        |             | C          |            | 8          | A        |                |                |                |                  |       |     |
|                       | 9         | 07/12/2017 |                 |               |         |                      |        | 0,014      | 41,5      | 84      | 0,5        | 0,8932      | C          |            |            | A        | 39             | 14.023         | 43.226         | 11               | 668   |     |
|                       | 9         | 21/05/2018 |                 |               |         |                      |        | 0,014      |           | 58      | 0,5        |             | C          |            | 11         | B        |                |                |                |                  |       |     |
|                       | 9         | 28/06/2018 |                 |               |         |                      |        | 0,014      | 41,0      | 85      | 0,5        | 0,8805      | C          |            | 8          | A        | 18             | 11.431         | 48.283         | 13               | 700   |     |
|                       | 9         | 24/07/2018 |                 |               |         |                      |        | 0,014      |           | 77      | 0,5        |             | C          |            | 8          | A        |                |                |                |                  |       |     |
|                       | 9         | 25/08/2018 |                 |               |         |                      |        | 0,017      |           | 82      | 0,5        |             | C          |            | 7          | A        |                |                |                |                  |       |     |
|                       | 9         | 25/09/2018 |                 |               |         |                      |        | 0,017      |           | 83      | 0,5        |             | C          |            | 6          | A        |                |                |                |                  |       |     |
|                       | 9         | 23/10/2018 |                 |               |         |                      |        | 0,017      |           | 78      | 0,5        |             | C          |            | 8          | A        |                |                |                |                  |       |     |
| 7                     | 3         | 27/02/17   | Trafo UAT       | 3             | 1035082 | 2010                 | 4.749  | Shandong   | 0,017     | 37,6    | 94         | 0,5         | 0,8768     | C          |            | 9        | A              | 20             | 12.535         | 56.879           | 20    | 985 |
|                       | 10        | 04/07/17   |                 |               |         |                      |        | 0,022      | 38,1      | 60      | 0,5        | 0,8767      | C          |            | 11         | A        | 20             | 16.028         | 68.308         | 20               | 971   |     |
|                       | 10        | 30/07/2017 |                 |               |         |                      |        | 0,025      |           | 88      | 0,5        |             | C          |            | 14         | B        |                |                |                |                  |       |     |
|                       | 10        | 30/08/2017 |                 |               |         |                      |        | 0,022      |           | 89      | 0,5        |             | C          |            | 11         | A        |                |                |                |                  |       |     |
|                       | 10        | 29/09/2017 |                 |               |         |                      |        | 0,020      |           | 80      | 0,5        |             | C          |            | 12         | A        |                |                |                |                  |       |     |





| no        | Test Cer. | Date Test  | nama unit   | MALAYSIAN LAB RECOMMENDATION |                                                                         | PAK SOETJAHJONO (TXM ADVISOR)<br>RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|-----------|------------|-------------|------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standards |           |            |             |                              |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           |           |            |             |                              |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 30/08/2017 |             |                              | Periodic condition monitoring                                           | <b>FOR TEST 28 Juni 2018</b><br><br>Masih ada beberapa trafo <b>ESP ( no 1 Unit 1 dan no 2 Unit 2)</b> yang Acid dan IFT di bawah standard. Kami sarankan untuk <b>ganti minyak.</b><br><br>Trafo-trafo <b>ESP (no 2,3,4 unit 1 , no 2,3,4 Unit 2 , n 2,3 Unit 3)</b> dan <b>Ash Handling 1</b> yang Grade Misture di bawah B, kami sarankan untuk dilakukan <b>Purifikasi</b><br><br>Pada <b>ESP 2,3,4 Unit 3</b> ditemukan Spesific Grafity di bawah stadanrd. Kemungkinan penggantian oli menggunakan <b>bukan Mineral Oli.</b> Mohon dicek kembali type oli pengganti atau dilakukan trending dalam 12 bulan mendatang.<br><br><b>FURAN</b><br><b>Semua Trafo ESP Unit 1, ESP 2 Unit 2 dan ESP 3 Unit 3</b> telah ditemukan jejak Furan akibat moisture tinggi yang cukup lama. Kami sarankan <b>untuk menjaga kondisi moisture</b> pada waktu selanjutnya agar perkembangan Furan tidak berlanjut<br><b>Main Trafo Unit 2</b> ditemukan jejak Furan 5M2F (ada hot spot) tetapi tidak ditemukan Combustible Gas berlebih pada test DGA. Hal ini mengakibatkan pengamanan terhadapat gangguan petir kurang memadai. Kami sarankan untuk <b>diperbaiki baiksystem groundingnya maupun arresternya</b><br><br><b>CORROSIVE SULPHUR</b><br><b>Main Trafo Unit 1 dan 3</b> serta <b>trafo UAT</b> dideteksi mulai ada corrosive sulphur. Untuk mencegah percepatan pertumbuhannya dan memperpanjang umur trafo, kami sarankan <b>menambah Passivator</b><br><br><b>FOR TEST 21 Mei 2018</b><br><br>Mengingat banyak trafo <b>ESP (Tc 5,6,7,8,11,12,13)</b> yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja. Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami saat penggantian minyak melalui purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg diatas 7 cycle<br><br>Trafo <b>ESP (Tc 6,7,8)</b> Furan (FOL) banyak berkurang karena moisture cukup banyak dalam kertas dan kemungkinan juga kualitas kertas kurang bagus<br><br>Untuk trafo <b>Ash Handling</b> sebaiknya segera dipurifikasi juga dengan temperatur diats 650c dan vacuum diatas 74 CMHg sekitar >7 cycle<br><br>Trafo <b>ESP 1 unit 2</b> , nilai Dielectric nya cukup rendah ditambah dengan kondisi Moisture yang banyak sebaiknya segera dilakukan |
|           | 7         | 29/09/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 30/10/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 27/11/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 07/12/2017 |             |                              | Retest 6 months: Acidity, IFT                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 21/05/2018 |             |                              | Retest 6 months: <b>Acidity, IFT, Dielectric and Water Content</b>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 28/06/2018 |             |                              | Add inhibitor 2,6-DBPC and PASSIVATOR                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 24/07/2018 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 25/08/2018 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 25/09/2018 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 7         | 23/10/2018 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5         | 8         | 04/07/17   | Trafo 4 ESP | 1                            | <b>Perform OIL REGENERATION</b><br>Inspect silica gel                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 30/07/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 30/08/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 29/09/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 30/10/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 27/11/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 07/12/2017 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 21/05/2018 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 28/06/2018 |             |                              | Perform <b>OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 24/07/2018 |             |                              | <b>Perform OIL REGENERATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 25/08/2018 |             |                              | Periodic condition monitoring (change oil on beginning of August 2018)  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 25/09/2018 |             |                              | <b>Perform OIL PURIFICATION</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 8         | 23/10/2018 |             |                              | <b>Perform OIL PURIFICTION</b>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6         | 2         | 27/02/17   | Main trafo  | 3                            | Retest 6 months: DGA                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           |           |            |             |                              | Other parameters: Periodic Condition Monitoring                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 04/07/17   |             |                              | Add Passivator                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           |           |            |             |                              | Periodic Condition Monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 30/07/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 30/08/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 29/09/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 30/10/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 27/11/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 07/12/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 21/05/2018 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 28/06/2018 |             |                              | Add passivator                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 24/07/2018 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 25/08/2018 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 25/09/2018 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 9         | 23/10/2018 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7         | 3         | 27/02/17   | Trafo UAT   | 3                            | Retest 6 months: DGA<br>Other parameters: Periodic Condition Monitoring |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 10        | 04/07/17   |             |                              | Add Passivator<br>Periodic Condition Monitoring<br>Inspect silica gel   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 10        | 30/07/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 10        | 30/08/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|           | 10        | 29/09/2017 |             |                              | Periodic condition monitoring                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



| no                    | Test Cer. | Date Test  | nama unit       | Serial number | thn opr   | Jumlah minyak, liter | Buatan | hasil test |           |         |            |             |            |            |            |          |                |                |                |                  |       |     |
|-----------------------|-----------|------------|-----------------|---------------|-----------|----------------------|--------|------------|-----------|---------|------------|-------------|------------|------------|------------|----------|----------------|----------------|----------------|------------------|-------|-----|
|                       |           |            |                 |               |           |                      |        | acid       | ift       | diel.   | color      | spec. Graf. | visual     | sedi.      | moisture   |          | dga ASTM       |                |                |                  |       |     |
|                       |           |            |                 |               |           |                      |        |            |           |         |            |             |            |            | ppm        | grade    | H <sub>2</sub> | O <sub>2</sub> | N <sub>2</sub> | CH <sub>4</sub>  | CO    |     |
| Standard yang dipakai |           |            |                 |               |           |                      |        | ASTM D974  | ASTM D971 | IEC 156 | ASTM D1500 | ASTM D4052  | ASTM D1524 | ASTM D1698 | ASTM D1533 | hydrogen | oksigen        | nitrogen       | methane        | carbon monoksida |       |     |
|                       |           |            | BATASAN OPERASI |               |           |                      |        | < 0,06     | > 32      | > 30    | < 3,5      | > 0,84      | C          |            | < 20       |          | < 100          |                |                | < 120            | < 350 |     |
|                       | 10        | 30/10/2017 |                 |               |           |                      |        | 0,022      |           | 86      | 0,5        |             | C          |            | 11         | A        |                |                |                |                  |       |     |
|                       | 10        | 27/11/2017 |                 |               |           |                      |        | 0,022      |           | 86      | 0,5        |             | C          |            | 12         | A        |                |                |                |                  |       |     |
|                       | 10        | 07/12/2017 |                 |               |           |                      |        | 0,020      | 37,6      | 86      | 0,5        | 0,8200      | C          |            | 12         | A        | 36             | 18.752         | 66.758         | 1                | 1.116 |     |
|                       | 10        | 21/05/2018 |                 |               |           |                      |        | 0,020      |           | 81      | 0,5        |             | C          |            | 14         | B        |                |                |                |                  |       |     |
|                       | 10        | 28/06/2018 |                 |               |           |                      |        | 0,022      | 37,5      | 79      | 0,5        | 0,8793      | C          |            | 10         | A        | 35             | 13.798         | 68.180         | 22               | 1.144 |     |
|                       | 10        | 24/07/2018 |                 |               |           |                      |        | 0,020      |           | 76      | 0,5        |             | C          |            | 11         | A        |                |                |                |                  |       |     |
|                       | 10        | 25/08/2018 |                 |               |           |                      |        | 0,022      |           | 78      | 0,5        |             | C          |            | 8          | A        |                |                |                |                  |       |     |
|                       | 10        | 25/09/2018 |                 |               |           |                      |        | 0,020      |           | 75      | 0,5        |             | C          |            | 8          | A        |                |                |                |                  |       |     |
|                       | 10        | 23/10/2018 |                 |               |           |                      |        | 0,020      |           | 69      | 0,5        |             | C          |            | 8          | A        |                |                |                |                  |       |     |
| 8                     | 11        | 04/07/17   | Trafo 1 ESP     | 3             | 007308428 | 2010                 | 731    | Zhejiang   | 0,022     | 23,2    | 32         | 1,5         | 0,8432     | C          |            | 26       | D              | 14             | 12.943         | 68.807           | 3     | 62  |
|                       | 11        | 30/07/2017 |                 |               |           |                      |        | 0,028      |           | 47      | 1,5        |             | C          |            | 21         | D        |                |                |                |                  |       |     |
|                       | 11        | 30/08/2017 |                 |               |           |                      |        | 0,028      |           | 36      | 1,5        |             | C          |            | 22         | D        |                |                |                |                  |       |     |
|                       | 11        | 29/09/2017 |                 |               |           |                      |        | 0,025      |           | 45      | 1,5        |             | C          |            | 20         | C        |                |                |                |                  |       |     |
|                       | 11        | 30/10/2017 |                 |               |           |                      |        | 0,025      |           | 40      | 1,5        |             | C          |            | 19         | D        |                |                |                |                  |       |     |
|                       | 11        | 27/11/2017 |                 |               |           |                      |        | 0,025      |           | 31      | 1,5        |             | C          |            | 34         | C        |                |                |                |                  |       |     |
|                       | 11        | 07/12/2017 |                 |               |           |                      |        | 0,022      | 34,6      | 31      | 1,5        | 0,8190      | C          |            | 31         | C        | 14             | 12.739         | 65.097         | 3                | 66    |     |
|                       | 11        | 21/05/2018 |                 |               |           |                      |        | 0,022      |           | 31      | 0,5        |             | C          |            | 27         | D        |                |                |                |                  |       |     |
|                       | 11        | 28/06/2018 |                 |               |           |                      |        | 0,022      | 35,0      | 60      | 0,5        | 0,8154      | C          |            | 10         | B        | 19             | 15.552         | 64.448         | 3                | 57    |     |
|                       | 11        | 24/07/2018 |                 |               |           |                      |        | 0,020      |           | 53      | 0,5        |             | C          |            | 9          | B        |                |                |                |                  |       |     |
|                       | 11        | 25/08/2018 |                 |               |           |                      |        | 0,022      |           | 77      | 0,5        |             | C          |            | 9          | B        |                |                |                |                  |       |     |
|                       | 11        | 25/09/2018 |                 |               |           |                      |        | 0,020      |           | 64      | 0,5        |             | C          |            | 10         | B        |                |                |                |                  |       |     |
|                       | 11        | 23/10/2018 |                 |               |           |                      |        | 0,020      |           | 63      | 0,5        |             | C          |            | 12         | B        |                |                |                |                  |       |     |
| 9                     | 12        | 04/07/17   | Trafo 2 ESP     | 3             | 007008428 | 2010                 | 731    | Zhejiang   | 0,118     | 16,2    | 24         | 3,0         | 0,8435     | C          |            | 39       | D              | 38             | 16.781         | 70.523           | 5     | 129 |
|                       | 12        | 30/07/2017 |                 |               |           |                      |        | 0,112      |           | 28      | 3,0        |             | C          |            | 38         | E        |                |                |                |                  |       |     |
|                       | 12        | 30/08/2017 |                 |               |           |                      |        | 0,109      |           | 34      | 2,5        |             | C          |            | 17         | D        |                |                |                |                  |       |     |
|                       | 12        | 29/09/2017 |                 |               |           |                      |        | 0,115      |           | 21      | 2,5        |             | C          |            | 37         | D        |                |                |                |                  |       |     |
|                       | 12        | 30/10/2017 |                 |               |           |                      |        | 0,118      |           | 33      | 2,5        |             | C          |            | 20         | D        |                |                |                |                  |       |     |
|                       | 12        | 27/11/2017 |                 |               |           |                      |        | 0,017      |           | 25      | 0,5        |             | C          |            | 28         | B        |                |                |                |                  |       |     |
|                       | 12        | 07/12/2017 |                 |               |           |                      |        | 0,022      | 29,7      | 29      | 0,1        | 0,8106      | C          |            | 44         | D        | 42             | 19.523         | 66.059         | 3                | 102   |     |
|                       | 12        | 21/05/2018 |                 |               |           |                      |        | 0,022      |           | 28      | 0,5        |             | C          |            | 30         | E        |                |                |                |                  |       |     |
|                       | 12        | 28/06/2018 |                 |               |           |                      |        | 0,025      | 30,2      | 34      | 0,5        | 0,8080      | C          |            | 16         | C        | 43             | 22.272         | 68.719         | 4                | 38    |     |
|                       | 12        | 24/07/2018 |                 |               |           |                      |        | 0,022      |           | 30      | 0,5        |             | C          |            | 25         | D        |                |                |                |                  |       |     |
|                       | 12        | 25/08/2018 |                 |               |           |                      |        | 0,022      |           | 67      | 0,5        |             | C          |            | 11         | C        |                |                |                |                  |       |     |
|                       | 12        | 25/09/2018 |                 |               |           |                      |        | 0,022      |           | 31      | 0,5        |             | C          |            | 32         | D        |                |                |                |                  |       |     |
|                       | 12        | 23/10/2018 |                 |               |           |                      |        | 0,025      |           | 54      | 0,5        |             | C          |            | 13         | B        |                |                |                |                  |       |     |
| 10                    | 13        | 04/07/17   | Trafo 3 ESP     | 3             | 007208428 | 2010                 | 731    | Zhejiang   | 0,025     | 27,8    | 40         | 1,5         | 0,8415     | C          |            | 27       | D              | 21             | 16.364         | 70.694           | 3     | 38  |
|                       | 13        | 30/07/2017 |                 |               |           |                      |        | 0,025      |           | 66      | 1,5        |             | C          |            | 18         | D        |                |                |                |                  |       |     |
|                       | 13        | 30/08/2017 |                 |               |           |                      |        | 0,025      |           | 50      | 1,5        |             | C          |            | 21         | C        |                |                |                |                  |       |     |
|                       | 13        | 29/09/2017 |                 |               |           |                      |        | 0,028      |           | 59      | 1,5        |             | C          |            | 22         | C        |                |                |                |                  |       |     |
|                       | 13        | 30/10/2017 |                 |               |           |                      |        | 0,028      |           | 55      | 1,5        |             | C          |            | 28         | D        |                |                |                |                  |       |     |
|                       | 13        | 27/11/2017 |                 |               |           |                      |        | 0,014      |           | 39      | 0,5        |             | C          |            | 24         | B        |                |                |                |                  |       |     |
|                       | 13        | 07/12/2017 |                 |               |           |                      |        | 0,014      | 39,8      | 45      | 0,5        | 0,8099      | C          |            | 29         | B        | 15             | 14.515         | 65.277         | 2                | 40    |     |
|                       | 13        | 21/05/2018 |                 |               |           |                      |        | 0,014      |           | 54      | 0,5        |             | C          |            | 23         | D        |                |                |                |                  |       |     |
|                       | 13        | 28/06/2018 |                 |               |           |                      |        | 0,017      | 38,4      | 43      | 0,5        | 0,8075      | C          |            | 22         | C        | 30             | 15.202         | 64.188         | 4                | 65    |     |



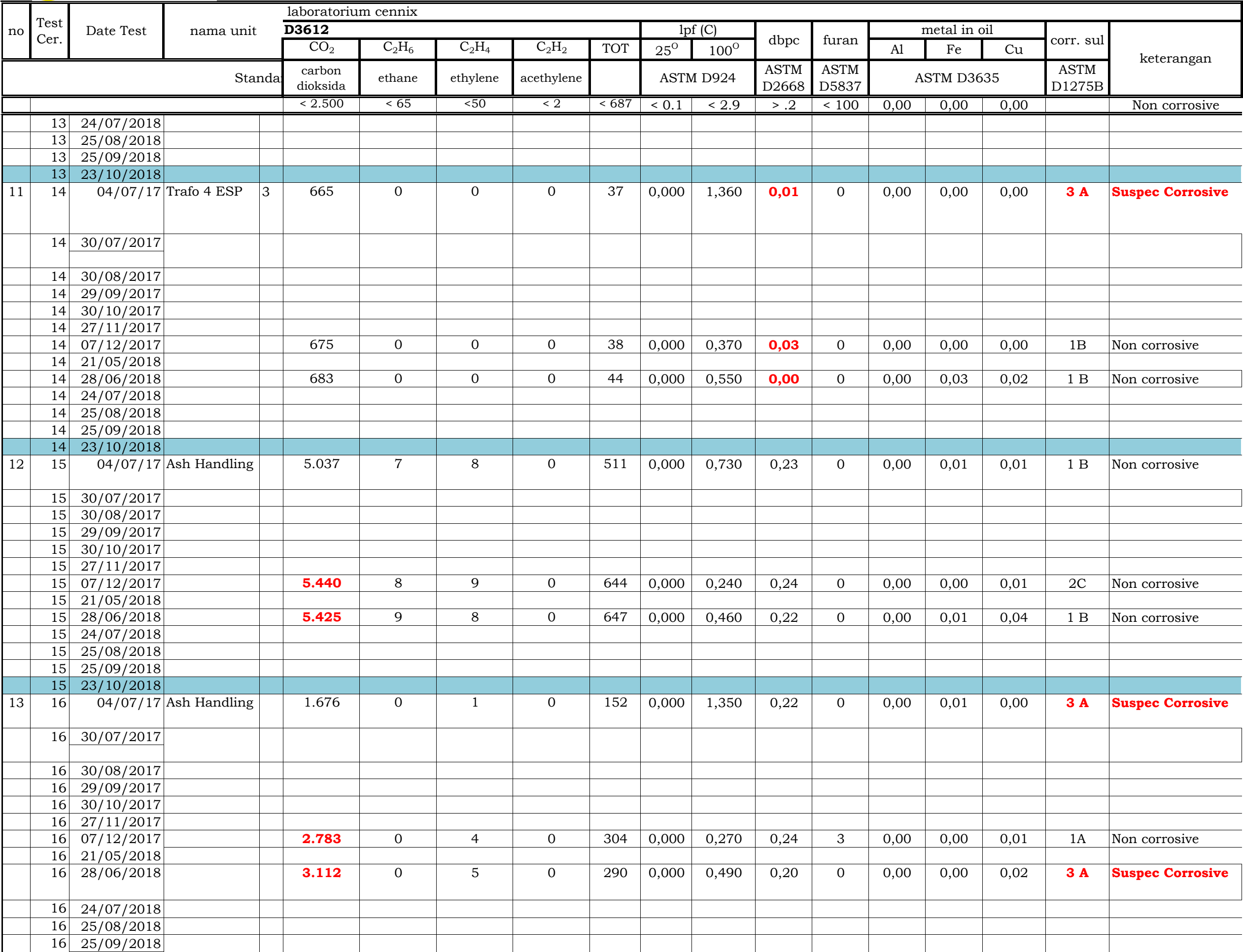
| no | Test Cer. | Date Test  | nama unit   |   | laboratorium cennix |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|----|-----------|------------|-------------|---|---------------------|-------------------------------|-------------------------------|-------------------------------|-------|-----------------|------------------|------|-------|--------------|------------|------|-----------|------------------|
|    |           |            |             |   | D3612               |                               |                               |                               |       | lpf (C)         |                  | dbpc | furan | metal in oil |            |      | corr. sul | keterangan       |
|    |           |            |             |   | CO <sub>2</sub>     | C <sub>2</sub> H <sub>6</sub> | C <sub>2</sub> H <sub>4</sub> | C <sub>2</sub> H <sub>2</sub> | TOT   | 25 <sup>o</sup> | 100 <sup>o</sup> |      |       | Al           | Fe         | Cu   |           |                  |
|    |           |            |             |   | carbon dioksida     | ethane                        | ethylene                      | acethylene                    |       | ASTM D924       | ASTM D2668       |      |       | ASTM D5837   | ASTM D3635 |      |           |                  |
|    |           |            | Standar     |   | < 2.500             | < 65                          | <50                           | < 2                           | < 687 | < 0.1           | < 2.9            | > .2 | < 100 | 0,00         | 0,00       | 0,00 |           | Non corrosive    |
|    | 10        | 30/10/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 10        | 27/11/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 10        | 07/12/2017 |             |   | 3.622               | 5                             | 7                             | 0                             | 1.165 | 0,000           | 0,170            | 0,23 | 0     | 0,00         | 0,01       | 0,08 | 2C        | Non corrosive    |
|    | 10        | 21/05/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 10        | 28/06/2018 |             |   | 3.644               | 5                             | 7                             | 1                             | 1.214 | 0,000           | 0,140            | 0,21 | 0     | 0,00         | 0,01       | 0,11 | 3 A       | Suspec Corrosive |
|    | 10        | 24/07/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 10        | 25/08/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 10        | 25/09/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 10        | 23/10/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
| 8  | 11        | 04/07/17   | Trafo 1 ESP | 3 | 1.243               | 0                             | 0                             | 0                             | 79    | 0,000           | 0,350            | 0,00 | 0     | 0,01         | 0,01       | 0,00 | 1 B       | Non corrosive    |
|    | 11        | 30/07/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 11        | 30/08/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 11        | 29/09/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 11        | 30/10/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 11        | 27/11/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 11        | 07/12/2017 |             |   | 612                 | 0                             | 0                             | 0                             | 83    | 0,000           | 0,040            | 0,18 | 0     | .01          | 0,00       | 0,00 | 2E        | Non corrosive    |
|    | 11        | 21/05/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 11        | 28/06/2018 |             |   | 1.391               | 0                             | 0                             | 0                             | 79    | 0,000           | 0,050            | 0,16 | 0     | 0,00         | 0,00       | 0,02 | 1 B       | Non corrosive    |
|    | 11        | 24/07/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 11        | 25/08/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 11        | 25/09/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 11        | 23/10/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
| 9  | 12        | 04/07/17   | Trafo 2 ESP | 3 | 2.431               | 3                             | 5                             | 0                             | 180   | 0,000           | 1,330            | 0,00 | 169   | 0,00         | 0,01       | 0,01 | 2 C       | Non corrosive    |
|    | 12        | 30/07/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 12        | 30/08/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 12        | 29/09/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 12        | 30/10/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 12        | 27/11/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 12        | 07/12/2017 |             |   | 912                 | 0                             | 0                             | 0                             | 147   | 0,000           | 0,030            | 0,20 | 0     | 0,00         | 0,00       | 0,00 | 3A        | Non corrosive    |
|    | 12        | 21/05/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 12        | 28/06/2018 |             |   | 1.226               | 0                             | 0                             | 0                             | 85    | 0,000           | 0,040            | 0,19 | 0     | 0,00         | 0,00       | 0,02 | 3 A       | Suspec Corrosive |
|    | 12        | 24/07/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 12        | 25/08/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 12        | 25/09/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 12        | 23/10/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
| 10 | 13        | 04/07/17   | Trafo 3 ESP | 3 | 1.068               | 0                             | 0                             | 0                             | 62    | 0,000           | 0,620            | 0,01 | 0     | 0,00         | 0,01       | 0,00 | 2 C       | Non corrosive    |
|    | 13        | 30/07/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 13        | 30/08/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 13        | 29/09/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 13        | 30/10/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 13        | 27/11/2017 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 13        | 07/12/2017 |             |   | 559                 | 0                             | 0                             | 0                             | 57    | 0,000           | .08              | .21  | 0     | 0,00         | 0,00       | 0,00 | 2E        | Non corrosive    |
|    | 13        | 21/05/2018 |             |   |                     |                               |                               |                               |       |                 |                  |      |       |              |            |      |           |                  |
|    | 13        | 28/06/2018 |             |   | 1.833               | 0                             | 0                             | 0                             | 99    | 0,000           | 0,010            | 0,20 | 4     | 0,00         | 0,00       | 0,02 | 2 E       | Non corrosive    |



| no      | Test Cer. | Date Test  | nama unit   | MALAYSIAN LAB RECOMMENDATION |                                                                                                       | PAK SOETJAHJONO (TXM ADVISOR)<br>RECOMMENDATION                                                                                                                                                                                                                                                                                                                                 |
|---------|-----------|------------|-------------|------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standar |           |            |             |                              |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                 |
|         |           |            |             |                              |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 10        | 30/10/2017 |             |                              | Periodic condition monitoring                                                                         | dengan kondisi moisture yang buruk, sebaiknya segera melakukan Purifikasi.<br><br>Kondisi semua trafo <b>ESP Unit 2 (Tc 17, 18, 19, 20)</b> Grade Moisture rendah, yang artinya kandungan air dalam isolasi kertas cukup tinggi. Kami sarankan untuk dimonitor dengan test DGA agar bisa diantisipasi jika ada kondisi gangguan thermal yang terjadi pada trafo trafo tersebut. |
|         | 10        | 27/11/2017 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 10        | 07/12/2017 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 10        | 21/05/2018 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 10        | 28/06/2018 |             |                              | Retest 6 months: DGA<br>Add Passivator                                                                |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 10        | 24/07/2018 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 10        | 25/08/2018 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 10        | 25/09/2018 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 10        | 23/10/2018 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
| 8       | 11        | 04/07/17   | Trafo 1 ESP | 3                            | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 30/07/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 30/08/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 29/09/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 30/10/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 27/11/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 07/12/2017 |             |                              | Perform OIL PURIFICATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 21/05/2018 |             |                              | Perform OIL PURIFICATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 28/06/2018 |             |                              | Retest 12 months: Specific Gravity<br>Add Passivator                                                  |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 24/07/2018 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 25/08/2018 |             |                              | Periodic condition monitoring (Purifying on end of July 2018)                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 25/09/2018 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 11        | 23/10/2018 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
| 9       | 12        | 04/07/17   | Trafo 2 ESP | 3                            | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 30/07/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 30/08/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 29/09/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 30/10/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 27/11/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 07/12/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 21/05/2018 |             |                              | Perform OIL PURIFICATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 28/06/2018 |             |                              | Retest 12 months: Specific Gravity<br>Add Passivator                                                  |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 24/07/2018 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 25/08/2018 |             |                              | Periodic condition monitoring (Purifying on end of July 2018)                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 25/09/2018 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 12        | 23/10/2018 |             |                              | Periodic condition monitoring                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
| 10      | 13        | 04/07/17   | Trafo 3 ESP | 3                            | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 13        | 30/07/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 13        | 30/08/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 13        | 29/09/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 13        | 30/10/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 13        | 27/11/2017 |             |                              | Perform OIL REGENERATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 13        | 07/12/2017 |             |                              | Perform OIL PURIFICATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 13        | 21/05/2018 |             |                              | Perform OIL PURIFICATION                                                                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|         | 13        | 28/06/2018 |             |                              | Retest 12 months: Specific Gravity<br>Retest 6 months: Dielectric and Water Content<br>Add Passivator |                                                                                                                                                                                                                                                                                                                                                                                 |

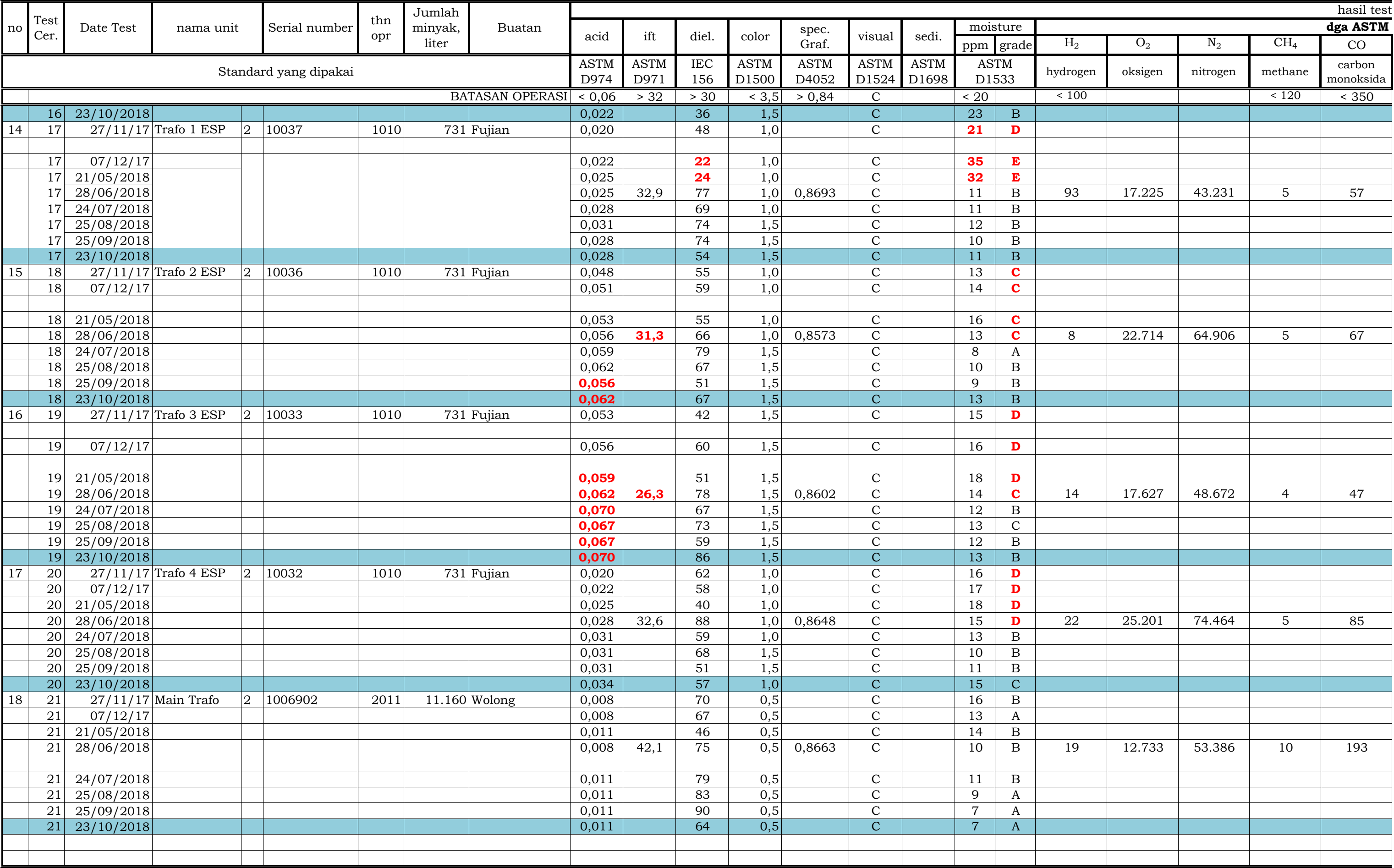


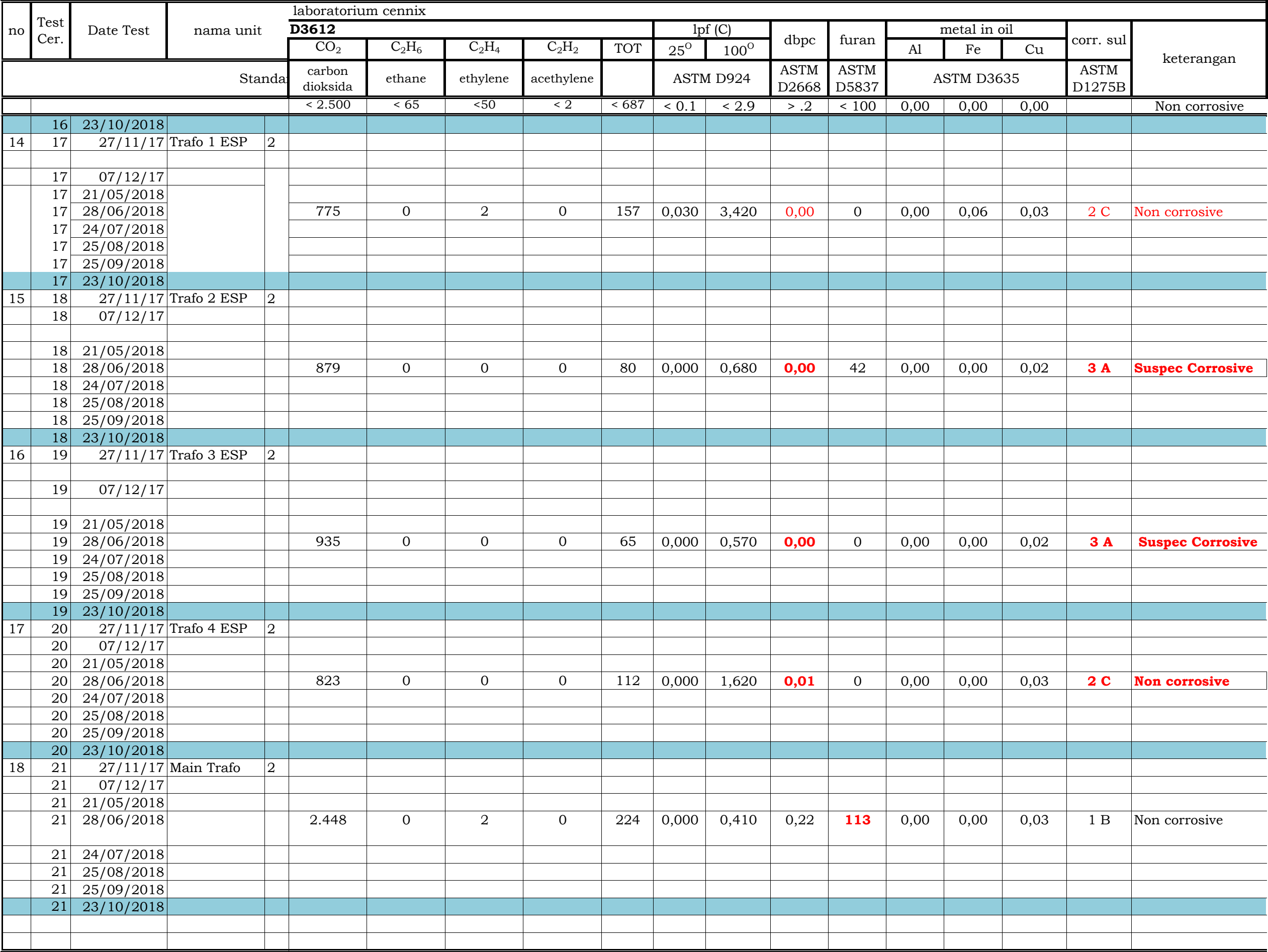
| no                    | Test Cer. | Date Test       | nama unit    | Serial number | thn opr   | Jumlah minyak, liter | Buatan | hasil test |           |         |            |             |            |            |            |          |                |                |                |                  |       |     |
|-----------------------|-----------|-----------------|--------------|---------------|-----------|----------------------|--------|------------|-----------|---------|------------|-------------|------------|------------|------------|----------|----------------|----------------|----------------|------------------|-------|-----|
|                       |           |                 |              |               |           |                      |        | acid       | ift       | diel.   | color      | spec. Graf. | visual     | sedi.      | moisture   |          | dga ASTM       |                |                |                  |       |     |
|                       |           |                 |              |               |           |                      |        |            |           |         |            |             |            |            | ppm        | grade    | H <sub>2</sub> | O <sub>2</sub> | N <sub>2</sub> | CH <sub>4</sub>  | CO    |     |
| Standard yang dipakai |           |                 |              |               |           |                      |        | ASTM D974  | ASTM D971 | IEC 156 | ASTM D1500 | ASTM D4052  | ASTM D1524 | ASTM D1698 | ASTM D1533 | hydrogen | oksigen        | nitrogen       | methane        | carbon monoksida |       |     |
|                       |           | BATASAN OPERASI |              |               |           |                      |        | < 0,06     | > 32      | > 30    | < 3,5      | > 0,84      | C          |            | < 20       |          | < 100          |                |                | < 120            | < 350 |     |
|                       | 13        | 24/07/2018      |              |               |           |                      |        | 0,014      |           | 43      | 0,5        |             | C          |            | 20         | C        |                |                |                |                  |       |     |
|                       | 13        | 25/08/2018      |              |               |           |                      |        | 0,014      |           | 76      | 0,5        |             | C          |            | 11         | C        |                |                |                |                  |       |     |
|                       | 13        | 25/09/2018      |              |               |           |                      |        | 0,020      |           | 47      | 0,5        |             | C          |            | 20         | B        |                |                |                |                  |       |     |
|                       | 13        | 23/10/2018      |              |               |           |                      |        | 0,014      |           | 49      | 0,5        |             | C          |            | 15         | A        |                |                |                |                  |       |     |
| 11                    | 14        | 04/07/17        | Trafo 4 ESP  | 3             | 007108428 | 2010                 | 731    | Zhejian    | 0,020     | 31,2    | 62         | 1,0         | 0,8420     | C          |            | 19       | D              | 10             | 18.669         | 66.036           | 3     | 24  |
|                       | 14        | 30/07/2017      |              |               |           |                      |        |            | 0,022     |         | 60         | 1,0         |            | C          |            | 15       | C              |                |                |                  |       |     |
|                       | 14        | 30/08/2017      |              |               |           |                      |        |            | 0,020     |         | 55         | 1,0         |            | C          |            | 17       | D              |                |                |                  |       |     |
|                       | 14        | 29/09/2017      |              |               |           |                      |        |            | 0,022     |         | 59         | 1,0         |            | C          |            | 15       | C              |                |                |                  |       |     |
|                       | 14        | 30/10/2017      |              |               |           |                      |        |            | 0,025     |         | 55         | 1,0         |            | C          |            | 15       | D              |                |                |                  |       |     |
|                       | 14        | 27/11/2017      |              |               |           |                      |        |            | 0,022     |         | 69         | 1,0         |            | C          |            | 16       | C              |                |                |                  |       |     |
|                       | 14        | 07/12/2017      |              |               |           |                      |        |            | 0,025     | 29,8    | 43         | 1,0         | 0,8471     | C          |            | 19       | D              | 14             | 15.414         | 63.712           | 2     | 22  |
|                       | 14        | 21/05/2018      |              |               |           |                      |        |            | 0,028     |         | 37         | 1,0         |            | C          |            | 21       | D              |                |                |                  |       |     |
|                       | 14        | 28/06/2018      |              |               |           |                      |        |            | 0,025     | 30,5    | 71         | 1,0         | 0,8442     | C          |            | 11       | A              | 20             | 14.690         | 67.822           | 2     | 22  |
|                       | 14        | 24/07/2018      |              |               |           |                      |        |            | 0,022     |         | 42         | 1,0         |            | C          |            | 11       | C              |                |                |                  |       |     |
|                       | 14        | 25/08/2018      |              |               |           |                      |        |            | 0,025     |         | 66         | 1,5         |            | C          |            | 11       | B              |                |                |                  |       |     |
|                       | 14        | 25/09/2018      |              |               |           |                      |        |            | 0,022     |         | 51         | 1,5         |            | C          |            | 11       | B              |                |                |                  |       |     |
|                       | 14        | 23/10/2018      |              |               |           |                      |        |            | 0,022     |         | 56         | 1,5         |            | C          |            | 12       | B              |                |                |                  |       |     |
| 12                    | 15        | 04/07/17        | Ash Handling |               | 10820108  | 2011                 | 1.023  | Wolong     | 0,022     | 38,1    | 38         | 1,0         | 0,8779     | C          |            | 28       | C              | 35             | 10.984         | 71.058           | 20    | 441 |
|                       | 15        | 30/07/2017      |              |               |           |                      |        |            | 0,025     |         | 56         | 1,0         |            | C          |            | 21       | D              |                |                |                  |       |     |
|                       | 15        | 30/08/2017      |              |               |           |                      |        |            | 0,022     |         | 53         | 1,0         |            | C          |            | 16       | B              |                |                |                  |       |     |
|                       | 15        | 29/09/2017      |              |               |           |                      |        |            | 0,022     |         | 57         | 1,0         |            | C          |            | 16       | B              |                |                |                  |       |     |
|                       | 15        | 30/10/2017      |              |               |           |                      |        |            | 0,022     |         | 45         | 1,0         |            | C          |            | 20       | B              |                |                |                  |       |     |
|                       | 15        | 27/11/2017      |              |               |           |                      |        |            | 0,020     |         | 36         | 1,0         |            | C          |            | 29       | C              |                |                |                  |       |     |
|                       | 15        | 07/12/2017      |              |               |           |                      |        |            | 0,017     | 37,5    | 34         | 1,0         | 0,8932     | C          |            | 26       | B              | 49             | 12.034         | 67.124           | 20    | 558 |
|                       | 15        | 21/05/2018      |              |               |           |                      |        |            | 0,022     |         | 33         | 1,0         |            | C          |            | 26       | C              |                |                |                  |       |     |
|                       | 15        | 28/06/2018      |              |               |           |                      |        |            | 0,025     | 37,5    | 42         | 1,0         | 0,8804     | C          |            | 24       | C              | 53             | 10.939         | 69.394           | 20    | 557 |
|                       | 15        | 24/07/2018      |              |               |           |                      |        |            | 0,022     |         | 39         | 1,0         |            | C          |            | 20       | B              |                |                |                  |       |     |
|                       | 15        | 25/08/2018      |              |               |           |                      |        |            | 0,025     |         | 60         | 1,5         |            | C          |            | 16       | B              |                |                |                  |       |     |
|                       | 15        | 25/09/2018      |              |               |           |                      |        |            | 0,025     |         | 43         | 1,5         |            | C          |            | 17       | A              |                |                |                  |       |     |
|                       | 15        | 23/10/2018      |              |               |           |                      |        |            | 0,022     |         | 28         | 1,5         |            | C          |            | 19       | C              |                |                |                  |       |     |
| 13                    | 16        | 04/07/17        | Ash Handling |               | 10820107  | 2011                 | 1.023  | Wolong     | 0,022     | 37,3    | 36         | 1,0         | 0,8788     | C          |            | 21       | C              | 18             | 21.157         | 64.224           | 4     | 129 |
|                       | 16        | 30/07/2017      |              |               |           |                      |        |            | 0,022     |         | 40         | 1,0         |            | C          |            | 32       | D              |                |                |                  |       |     |
|                       | 16        | 30/08/2017      |              |               |           |                      |        |            | 0,022     |         | 31         | 1,0         |            | C          |            | 29       | C              |                |                |                  |       |     |
|                       | 16        | 29/09/2017      |              |               |           |                      |        |            | 0,022     |         | 28         | 1,5         |            | C          |            | 31       | B              |                |                |                  |       |     |
|                       | 16        | 30/10/2017      |              |               |           |                      |        |            | 0,020     |         | 46         | 1,5         |            | C          |            | 21       | C              |                |                |                  |       |     |
|                       | 16        | 27/11/2017      |              |               |           |                      |        |            | 0,022     |         | 47         | 1,5         |            | C          |            | 20       | C              |                |                |                  |       |     |
|                       | 16        | 07/12/2017      |              |               |           |                      |        |            | 0,020     | 36,6    | 24         | 1,5         | 0,8940     | C          |            | 32       | D              | 20             | 15.791         | 63.530           | 5     | 275 |
|                       | 16        | 21/05/2018      |              |               |           |                      |        |            | 0,020     |         | 24         | 1,5         |            | C          |            | 32       | D              |                |                |                  |       |     |
|                       | 16        | 28/06/2018      |              |               |           |                      |        |            | 0,022     | 35,0    | 50         | 1,5         | 0,8811     | C          |            | 16       | B              | 32             | 22.904         | 67.139           | 4     | 249 |
|                       | 16        | 24/07/2018      |              |               |           |                      |        |            | 0,020     |         | 27         | 1,5         |            | C          |            | 30       | C              |                |                |                  |       |     |
|                       | 16        | 25/08/2018      |              |               |           |                      |        |            | 0,022     |         | 45         | 1,5         |            | C          |            | 16       | B              |                |                |                  |       |     |
|                       | 16        | 25/09/2018      |              |               |           |                      |        |            | 0,022     |         | 36         | 1,5         |            | C          |            | 19       | B              |                |                |                  |       |     |





| no      | Test Cer. | Date Test  | nama unit    | MALAYSIAN LAB RECOMMENDATION |                                                                                                                                   | PAK SOETJAHJONO (TXM ADVISOR)<br>RECOMMENDATION |
|---------|-----------|------------|--------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Standar |           |            |              |                              |                                                                                                                                   |                                                 |
|         |           |            |              |                              |                                                                                                                                   |                                                 |
|         | 13        | 24/07/2018 |              |                              | <b>Perform OIL PURIFICATION</b>                                                                                                   |                                                 |
|         | 13        | 25/08/2018 |              |                              | Periodic condition monitoring (Purifying on end of July 2018)                                                                     |                                                 |
|         | 13        | 25/09/2018 |              |                              | <b>Perform OIL PURIFICATION</b>                                                                                                   |                                                 |
|         | 13        | 23/10/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
| 11      | 14        | 04/07/17   | Trafo 4 ESP  | 3                            | Retest 6 months: Acidity, IFT and LPF<br>Add inhibitor 2,6-DBPC and Passivator<br>Other parameters: Periodic Condition Monitoring |                                                 |
|         | 14        | 30/07/2017 |              |                              | Retest 6 months: <b>Acidity, IFT</b> and <b>Liquid Power Factor</b><br>Other parameters: Periodic Condition Monitoring            |                                                 |
|         | 14        | 30/08/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 14        | 29/09/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 14        | 30/10/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 14        | 27/11/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 14        | 07/12/2017 |              |                              | Retest 6 months: Acidity, IFT                                                                                                     |                                                 |
|         | 14        | 21/05/2018 |              |                              | Retest 6 months: <b>Dielectric and Water Content</b>                                                                              |                                                 |
|         | 14        | 28/06/2018 |              |                              | Add inhibitor 2,6-DBPC and PASSIVATOR                                                                                             |                                                 |
|         | 14        | 24/07/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 14        | 25/08/2018 |              |                              | Periodic condition monitoring (Purifying on end of July 2018)                                                                     |                                                 |
|         | 14        | 25/09/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 14        | 23/10/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
| 12      | 15        | 04/07/17   | Ash Handling |                              | Add Passivator<br>Periodic Condition Monitoring                                                                                   |                                                 |
|         | 15        | 30/07/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 30/08/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 29/09/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 30/10/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 27/11/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 07/12/2017 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 21/05/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 28/06/2018 |              |                              | Add passivator                                                                                                                    |                                                 |
|         | 15        | 24/07/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 25/08/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 25/09/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
|         | 15        | 23/10/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |
| 13      | 16        | 04/07/17   | Ash Handling |                              | Add Passivator<br>Periodic Condition Monitoring                                                                                   |                                                 |
|         | 16        | 30/07/2017 |              |                              | Retest 6 months: <b>Dielectric</b> and <b>Water Content</b><br>Other parameters: Periodic Condition Monitoring                    |                                                 |
|         | 16        | 30/08/2017 |              |                              | Periodic Condition Monitoring                                                                                                     |                                                 |
|         | 16        | 29/09/2017 |              |                              | Retest 6 months: <b>Dielectric</b> and <b>Water Content</b>                                                                       |                                                 |
|         | 16        | 30/10/2017 |              |                              | Retest 6 months: <b>Dielectric</b> and <b>Water Content</b>                                                                       |                                                 |
|         | 16        | 27/11/2017 |              |                              | Periodic Condition Monitoring                                                                                                     |                                                 |
|         | 16        | 07/12/2017 |              |                              | <b>Perform OIL PURIFICATION</b>                                                                                                   |                                                 |
|         | 16        | 21/05/2018 |              |                              | <b>Perform OIL PURIFICATION</b>                                                                                                   |                                                 |
|         | 16        | 28/06/2018 |              |                              | Retest 6 months: DGA<br>Add Passivator                                                                                            |                                                 |
|         | 16        | 24/07/2018 |              |                              | <b>Perform OIL PURIFICATION</b>                                                                                                   |                                                 |
|         | 16        | 25/08/2018 |              |                              | Periodic condition monitoring (Purifying on end of July 2018)                                                                     |                                                 |
|         | 16        | 25/09/2018 |              |                              | Periodic condition monitoring                                                                                                     |                                                 |







| no      | Test Cer. | Date Test  | nama unit   | MALAYSIAN LAB RECOMMENDATION |                                                         | PAK SOETJAHJONO (TXM ADVISOR)<br>RECOMMENDATION |
|---------|-----------|------------|-------------|------------------------------|---------------------------------------------------------|-------------------------------------------------|
| Standar |           |            |             |                              |                                                         |                                                 |
|         |           |            |             |                              |                                                         |                                                 |
|         | 16        | 23/10/2018 |             |                              | Periodic condition monitoring                           |                                                 |
| 14      | 17        | 27/11/17   | Trafo 1 ESP | 2                            | Retest 6 months: Dielectric and Water Content           |                                                 |
|         |           |            |             |                              | Other parameters: Periodic Condition Monitoring         |                                                 |
|         | 17        | 07/12/17   |             |                              | <b>OIL PURIFICATION</b>                                 |                                                 |
|         | 17        | 21/05/2018 |             |                              | <b>OIL PURIFICATION</b>                                 |                                                 |
|         | 17        | 28/06/2018 |             |                              | <b>Add inhibitor 2,6-DBPC and PASSIVATOR</b>            |                                                 |
|         | 17        | 24/07/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 17        | 25/08/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 17        | 25/09/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 17        | 23/10/2018 |             |                              | Periodic condition monitoring                           |                                                 |
| 15      | 18        | 27/11/17   | Trafo 2 ESP | 2                            | Periodic condition monitoring                           |                                                 |
|         | 18        | 07/12/17   |             |                              | Retest 6 months: Acidity and IFT                        |                                                 |
|         |           |            |             |                              | Periodic condition monitoring                           |                                                 |
|         | 18        | 21/05/2018 |             |                              | Retest 6 months: <b>Acidity and IFT</b>                 |                                                 |
|         | 18        | 28/06/2018 |             |                              | Add inhibitor 2,6-DBPC and PASSIVATOR                   |                                                 |
|         | 18        | 24/07/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 18        | 25/08/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 18        | 25/09/2018 |             |                              | <b>Perform OIL REGENERATION</b>                         |                                                 |
|         | 18        | 23/10/2018 |             |                              | <b>Perform OIL REGENERATION</b>                         |                                                 |
| 16      | 19        | 27/11/17   | Trafo 3 ESP | 2                            | Retest 6 months: Acidity and IFT                        |                                                 |
|         |           |            |             |                              | Other parameters: Periodic Condition Monitoring         |                                                 |
|         | 19        | 07/12/17   |             |                              | Retest 6 months: Acidity and IFT                        |                                                 |
|         |           |            |             |                              | Other parameters: Periodic Condition Monitoring         |                                                 |
|         | 19        | 21/05/2018 |             |                              | Retest 6 months: <b>Acidity and IFT</b>                 |                                                 |
|         | 19        | 28/06/2018 |             |                              | <b>Perform OIL REGENERATION</b>                         |                                                 |
|         | 19        | 24/07/2018 |             |                              | <b>Perform OIL REGENERATION</b>                         |                                                 |
|         | 19        | 25/08/2018 |             |                              | <b>Perform OIL REGENERATION</b>                         |                                                 |
|         | 19        | 25/09/2018 |             |                              | <b>Perform OIL REGENERATION</b>                         |                                                 |
|         | 19        | 23/10/2018 |             |                              | <b>Perform OIL REGENERATION</b>                         |                                                 |
| 17      | 20        | 27/11/17   | Trafo 4 ESP | 2                            | Periodic condition monitoring                           |                                                 |
|         | 20        | 07/12/17   |             |                              | Periodic condition monitoring                           |                                                 |
|         | 20        | 21/05/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 20        | 28/06/2018 |             |                              | Add inhibitor 2,6-DBPC and PASSIVATOR                   |                                                 |
|         | 20        | 24/07/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 20        | 25/08/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 20        | 25/09/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 20        | 23/10/2018 |             |                              | Periodic condition monitoring                           |                                                 |
| 18      | 21        | 27/11/17   | Main Trafo  | 2                            | Periodic condition monitoring                           |                                                 |
|         | 21        | 07/12/17   |             |                              | Periodic condition monitoring                           |                                                 |
|         | 21        | 21/05/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 21        | 28/06/2018 |             |                              | Retest 6 months: <b>Furan and DGA</b><br>Add Passivator |                                                 |
|         | 21        | 24/07/2018 |             |                              | Periodic Condition monitoring                           |                                                 |
|         | 21        | 25/08/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 21        | 25/09/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         | 21        | 23/10/2018 |             |                              | Periodic condition monitoring                           |                                                 |
|         |           |            |             |                              |                                                         |                                                 |
|         |           |            |             |                              |                                                         |                                                 |